Numerical Optimization

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Theory of Constrained Optimization

The second part of this book is about minimizing functions subject to constraints on the variables. A general formulation for these problems is

$$\min_{x \in \mathbb{R}^n} f(x) \qquad \text{subject to} \quad \begin{cases} c_i(x) = 0, & i \in \mathcal{E}, \\ c_i(x) \geq 0, & i \in \mathcal{I}, \end{cases} \tag{12.1}$$

where f and the functions c_i are all smooth, real-valued functions on a subset of $\mathbb{R}^n$, and $\mathcal{I}$ and $\mathcal{E}$ are two finite sets of indices. As before, we call f the *objective* function, while c_i, $i \in \mathcal{E}$ are the *equality constraints* and c_i, $i \in \mathcal{I}$ are the *inequality constraints*. We define the *feasible set* Ω to be the set of points x that satisfy the constraints; that is,

$$\Omega = \{x \mid c_i(x) = 0, \quad i \in \mathcal{E}; \quad c_i(x) \geq 0, \quad i \in \mathcal{I}\}, \tag{12.2}$$

so that we can rewrite (12.1) more compactly as

$$\min_{x \in \Omega} \; f(x). \tag{12.3}$$

In this chapter we derive mathematical characterizations of the solutions of (12.3). Recall that for the unconstrained optimization problem of Chapter 2, we characterized solution points x^* in the following way:

> Necessary conditions: Local minima of unconstrained problems have $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ positive semidefinite.
>
> Sufficient conditions: Any point x^* at which $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is positive definite is a strong local minimizer of f.

Our aim in this chapter is to derive similar conditions to characterize the solutions of constrained optimization problems.

LOCAL AND GLOBAL SOLUTIONS

We have seen already that global solutions are difficult to find even when there are no constraints. The situation may be improved when we add constraints, since the feasible set might exclude many of the local minima and it may be comparatively easy to pick the global minimum from those that remain. However, constraints can also make things much more difficult. As an example, consider the problem

$$\min_{x \in \mathbb{R}^n} \|x\|_2^2, \quad \text{subject to } \|x\|_2^2 \geq 1.$$

Without the constraint, this is a convex quadratic problem with unique minimizer $x = 0$. When the constraint is added, any vector x with $\|x\|_2 = 1$ solves the problem. There are infinitely many such vectors (hence, infinitely many local minima) whenever $n \geq 2$.

A second example shows how addition of a constraint produces a large number of local solutions that do not form a connected set. Consider

$$\min (x_2 + 100)^2 + 0.01 x_1^2, \quad \text{subject to } x_2 - \cos x_1 \geq 0, \tag{12.4}$$

illustrated in Figure 12.1. Without the constraint, the problem has the unique solution $(-100, 0)$. With the constraint there are local solutions near the points

$$(x_1, x_2) = (k\pi, -1), \quad \text{for} \quad k = \pm 1, \pm 3, \pm 5, \ldots.$$

Definitions of the different types of local solutions are simple extensions of the corresponding definitions for the unconstrained case, except that now we restrict consideration to the *feasible* points in the neighborhood of x^*. We have the following definition.

> A vector x^* is a *local solution* of the problem (12.3) if $x^* \in \Omega$ and there is a neighborhood $\mathcal{N}$ of x^* such that $f(x) \geq f(x^*)$ for $x \in \mathcal{N} \cap \Omega$.

Similarly, we can make the following definitions:

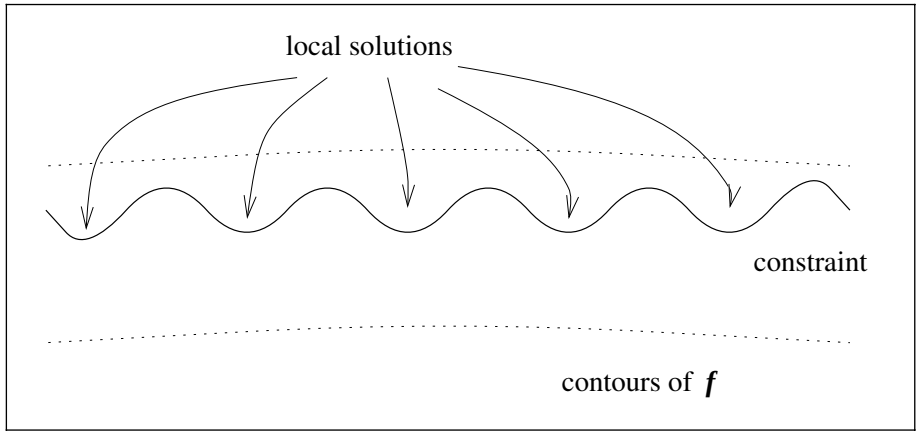


Figure 12.1 Constrained problem with many isolated local solutions.

A vector x^* is a *strict local solution* (also called a *strong local solution*) if $x^* \in \Omega$ and there is a neighborhood $\mathcal{N}$ of x^* such that $f(x) > f(x^*)$ for all $x \in \mathcal{N} \cap \Omega$ with $x \neq x^*$.

A point x^* is an *isolated local solution* if $x^* \in \Omega$ and there is a neighborhood $\mathcal{N}$ of x^* such that x^* is the only local minimizer in $\mathcal{N} \cap \Omega$.

At times, we replace the word "solution" by "minimizer" in our discussion. This alternative is frequently used in the literature, but it is slightly less satisfying because it does not account for the role of the constraints in defining the point in question.

SMOOTHNESS

Smoothness of objective functions and constraints is an important issue in characterizing solutions, just as in the unconstrained case. It ensures that the objective function and the constraints all behave in a reasonably predictable way and therefore allows algorithms to make good choices for search directions.

We saw in Chapter 2 that graphs of nonsmooth functions contain "kinks" or "jumps" where the smoothness breaks down. If we plot the feasible region for any given constrained optimization problem, we usually observe many kinks and sharp edges. Does this mean that the constraint functions that describe these regions are nonsmooth? The answer is often no, because the nonsmooth boundaries can often be described by a collection of smooth constraint functions. Figure 12.2 shows a diamond-shaped feasible region in $\mathbb{R}^2$ that could be described by the single nonsmooth constraint

$$\|x\|_1 = |x_1| + |x_2| \leq 1. \tag{12.5}$$

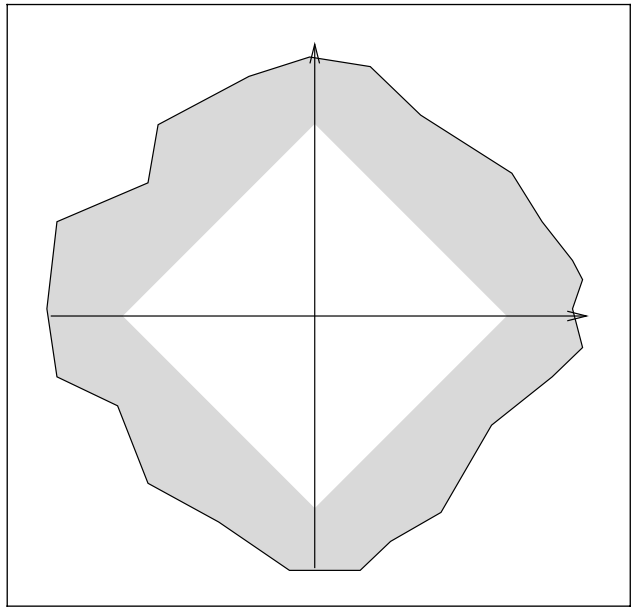


Figure 12.2 A feasible region with a nonsmooth boundary can be described by smooth constraints.

It can also be described by the following set of smooth (in fact, linear) constraints:

$$x_1 + x_2 \leq 1, \quad x_1 - x_2 \leq 1, \quad -x_1 + x_2 \leq 1, \quad -x_1 - x_2 \leq 1. \tag{12.6}$$

Each of the four constraints represents one edge of the feasible polytope. In general, the constraint functions are chosen so that each one represents a smooth piece of the boundary of Ω.

Nonsmooth, unconstrained optimization problems can sometimes be reformulated as smooth constrained problems. An example is given by the unconstrained scalar problem of minimizing a nonsmooth function $f(x)$ defined by

$$f(x) = \max(x^2, x), \tag{12.7}$$

which has kinks at $x = 0$ and $x = 1$, and the solution at $x^* = 0$. We obtain a smooth, constrained formulation of this problem by adding an artificial variable t and writing

$$\min t \qquad \text{s.t.} \qquad t \geq x, \quad t \geq x^2. \tag{12.8}$$

Reformulation techniques such as (12.6) and (12.8) are used often in cases where f is a maximum of a collection of functions or when f is a 1-norm or ∞-norm of a vector function.

In the examples above we expressed inequality constraints in a slightly different way from the form $c_i(x) \geq 0$ that appears in the definition (12.1). However, any collection of inequality constraints with $\geq$ and $\leq$ and nonzero right-hand-sides can be expressed in the form $c_i(x) \geq 0$ by simple rearrangement of the inequality. In general, it is good practice to state the constraint in a way that is intuitive and easy to understand.

12.1 EXAMPLES

To introduce the basic principles behind the characterization of solutions of constrained optimization problems, we work through three simple examples. The ideas discussed here will be made rigorous in the sections that follow.

We start by noting one item of terminology that recurs throughout the rest of the book: At a feasible point x, the inequality constraint $i \in \mathcal{I}$ is said to be *active* if $c_i(x) = 0$ and *inactive* if the strict inequality $c_i(x) > 0$ is satisfied.

A SINGLE EQUALITY CONSTRAINT

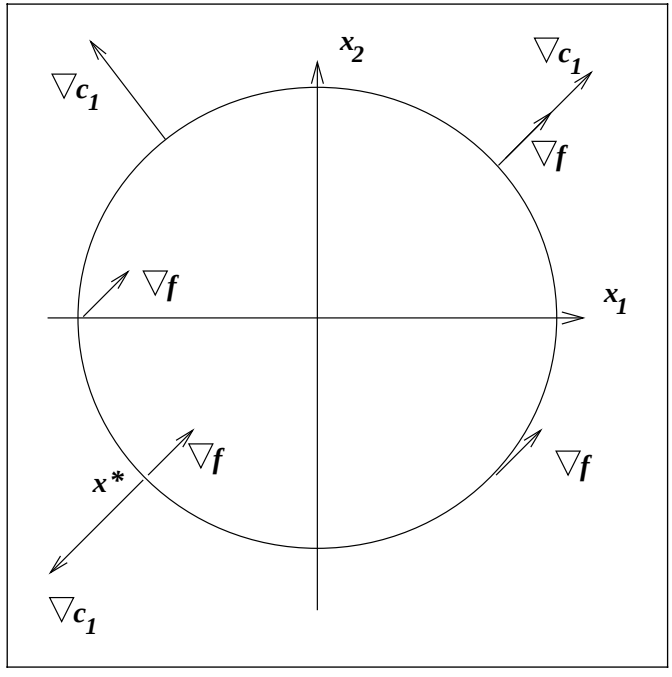


Figure 12.3 Problem (12.9), showing constraint and function gradients at various feasible points.

❑ EXAMPLE 12.1

Our first example is a two-variable problem with a single equality constraint:

$$\min\ x_1 + x_2 \qquad \text{s.t.} \qquad x_1^2 + x_2^2 - 2 = 0 \tag{12.9}$$

(see Figure 12.3). In the language of (12.1), we have $f(x) = x_1 + x_2$, $\mathcal{I} = \emptyset$, $\mathcal{E} = \{1\}$, and $c_1(x) = x_1^2 + x_2^2 - 2$. We can see by inspection that the feasible set for this problem is the circle of radius $\sqrt{2}$ centered at the origin—just the boundary of this circle, not its interior. The solution x^* is obviously $(-1, -1)^T$. From any other point on the circle, it is easy to find a way to move that *stays feasible* (that is, remains on the circle) while *decreasing* f. For instance, from the point $x = (\sqrt{2}, 0)^T$ any move in the clockwise direction around the circle has the desired effect.

We also see from Figure 12.3 that at the solution x^*, the *constraint normal* $\nabla c_1(x^*)$ is parallel to $\nabla f(x^*)$. That is, there is a scalar λ_1^* such that

$$\nabla f(x^*) = \lambda_1^* \nabla c_1(x^*). \tag{12.10}$$

(In this particular case, we have $\lambda_1^* = -\frac{1}{2}$.)



We can derive (12.10) by examining first-order Taylor series approximations to the objective and constraint functions. To retain feasibility with respect to the function $c_1(x) = 0$, we require that $c_1(x + d) = 0$; that is,

$$0 = c_1(x + d) \approx c_1(x) + \nabla c_1(x)^T d = \nabla c_1(x)^T d. \tag{12.11}$$

Hence, the direction d retains feasibility with respect to c_1, to first order, when it satisfies

$$\nabla c_1(x)^T d = 0. \tag{12.12}$$

Similarly, a direction of improvement must produce a decrease in f, so that

$$0 > f(x + d) - f(x) \approx \nabla f(x)^T d,$$

or, to first order,

$$\nabla f(x)^T d < 0. \tag{12.13}$$

If there exists a direction d that satisfies both (12.12) and (12.13), we conclude that improvement on our current point x is possible. It follows that a *necessary condition* for optimality for the problem (12.9) is that there exist *no direction d satisfying both (12.12) and (12.13).*

By drawing a picture, the reader can check that the only way that such a direction cannot exist is if $\nabla f(x)$ and $\nabla c_1(x)$ are parallel, that is, if the condition $\nabla f(x) = \lambda_1 \nabla c_1(x)$ holds at x, for some scalar λ_1. If this condition is *not* satisfied, the direction defined by

$$d = -\left(I - \frac{\nabla c_1(x)\nabla c_1(x)^T}{\|\nabla c_1(x)\|^2}\right)\nabla f(x) \tag{12.14}$$

satisfies both conditions (12.12) and (12.13) (see the exercises).

By introducing the *Lagrangian function*

$$\mathcal{L}(x, \lambda_1) = f(x) - \lambda_1 c_1(x), \tag{12.15}$$

and noting that $\nabla_x \mathcal{L}(x, \lambda_1) = \nabla f(x) - \lambda_1 \nabla c_1(x)$, we can state the condition (12.10) equivalently as follows: At the solution x^*, there is a scalar λ_1^* such that

$$\nabla_x \mathcal{L}(x^*, \lambda_1^*) = 0. \tag{12.16}$$

This observation suggests that we can search for solutions of the equality-constrained problem (12.9) by searching for stationary points of the Lagrangian function. The scalar quantity λ_1 in (12.15) is called a *Lagrange multiplier* for the constraint $c_1(x) = 0$.

Though the condition (12.10) (equivalently, (12.16)) appears to be *necessary* for an optimal solution of the problem (12.9), it is clearly not *sufficient*. For instance, in Example 12.1, (12.10) is satisfied at the point $x = (1, 1)$ (with $\lambda_1 = \frac{1}{2}$), but this point is obviously not a solution—in fact, it *maximizes* the function f on the circle. Moreover, in the case of equality-constrained problems, we cannot turn the condition (12.10) into a sufficient condition simply by placing some restriction on the sign of λ_1. To see this, consider replacing the constraint $x_1^2 + x_2^2 - 2 = 0$ by its negative $2 - x_1^2 - x_2^2 = 0$ in Example 12.1. The solution of the problem is not affected, but the value of λ_1^* that satisfies the condition (12.10) changes from $\lambda_1^* = -\frac{1}{2}$ to $\lambda_1^* = \frac{1}{2}$.

A SINGLE INEQUALITY CONSTRAINT

❑ EXAMPLE 12.2

This is a slight modification of Example 12.1, in which the equality constraint is replaced by an inequality. Consider

$$\min\ x_1 + x_2 \qquad \text{s.t.} \qquad 2 - x_1^2 - x_2^2 \geq 0, \tag{12.17}$$

for which the feasible region consists of the circle of problem (12.9) and its interior (see Figure 12.4). Note that the constraint normal ∇c_1 points toward the interior of the feasible

region at each point on the boundary of the circle. By inspection, we see that the solution is still $(-1, -1)$ and that the condition (12.10) holds for the value $\lambda_1^* = \frac{1}{2}$. However, this inequality-constrained problem differs from the equality-constrained problem (12.9) of Example 12.1 in that the sign of the Lagrange multiplier plays a significant role, as we now argue. ◻

As before, we conjecture that a given feasible point x is *not* optimal if we can find a step d that both retains feasibility and decreases the objective function f to first order. The main difference between problems (12.9) and (12.17) comes in the handling of the feasibility condition. As in (12.13), the direction d improves the objective function, to first order, if $\nabla f(x)^T d < 0$. Meanwhile, the direction d retains feasibility if

$$0 \leq c_1(x + d) \approx c_1(x) + \nabla c_1(x)^T d,$$

so, to first order, feasibility is retained if

$$c_1(x) + \nabla c_1(x)^T d \geq 0. \tag{12.18}$$

In determining whether a direction d exists that satisfies both (12.13) and (12.18), we consider the following two cases, which are illustrated in Figure 12.4.

Case I: Consider first the case in which x lies *strictly inside* the circle, so that the strict inequality $c_1(x) > 0$ holds. In this case, *any* vector d satisfies the condition (12.18), provided only that its length is sufficiently small. In particular, whenever $\nabla f(x^*) \neq 0$, we can obtain a direction d that satisfies both (12.13) and (12.18) by setting

$$d = -c_1(x) \frac{\nabla f(x)}{\|\nabla f(x)\|}.$$

The only situation in which such a direction fails to exist is when

$$\nabla f(x) = 0. \tag{12.19}$$

Case II: Consider now the case in which x lies on the boundary of the circle, so that $c_1(x) = 0$. The conditions (12.13) and (12.18) therefore become

$$\nabla f(x)^T d < 0, \quad \nabla c_1(x)^T d \geq 0.$$

The first of these conditions defines an open half-space, while the second defines a closed half-space, as illustrated in Figure 12.5. It is clear from this figure that the two regions fail to intersect only when $\nabla f(x)$ and $\nabla c_1(x)$ point in the same direction, that is, when

$$\nabla f(x) = \lambda_1 \nabla c_1(x), \quad \text{for some } \lambda_1 \geq 0. \tag{12.20}$$

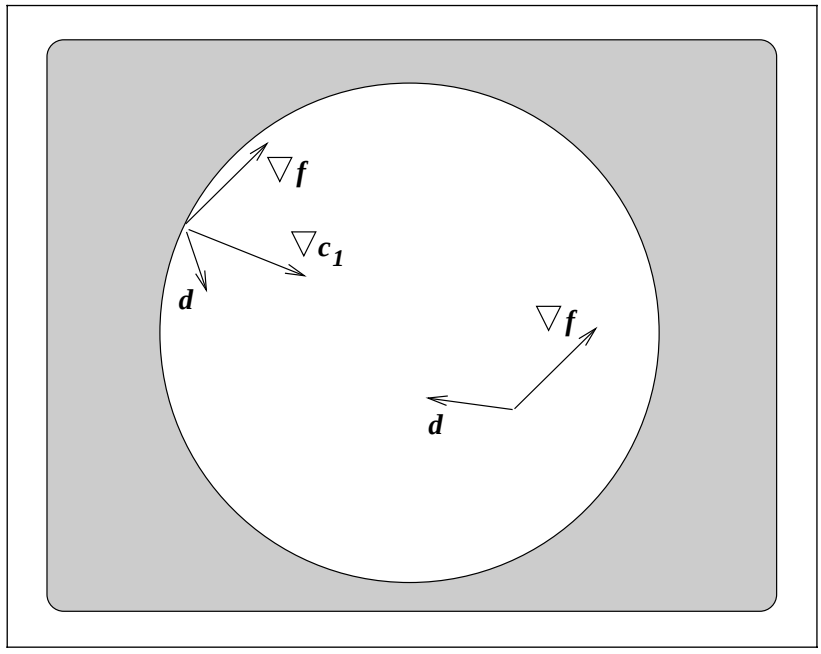


Figure 12.4 Improvement directions from two feasible points for the problem (12.17) at which the constraint is active and inactive, respectively.

Note that the sign of the multiplier is significant here. If (12.10) were satisfied with a *negative* value of λ_1, then $\nabla f(x)$ and $\nabla c_1(x)$ would point in opposite directions, and we see from Figure 12.5 that the set of directions that satisfy both (12.13) and (12.18) would make up an entire open half-plane.

The optimality conditions for both cases I and II can again be summarized neatly with reference to the Lagrangian function. When no first-order feasible descent direction exists at some point x^*, we have that

$$\nabla_x \mathcal{L}(x^*, \lambda_1^*) = 0, \quad \text{for some } \lambda_1^* \geq 0, \tag{12.21}$$

where we also require that

$$\lambda_1^* c_1(x^*) = 0. \tag{12.22}$$

This condition is known as a *complementarity condition*; it implies that the Lagrange multiplier λ_1 can be strictly positive *only when the corresponding constraint c_1 is active*. Conditions of this type play a central role in constrained optimization, as we see in the sections that follow. In case I, we have that $c_1(x^*) > 0$, so (12.22) requires that $\lambda_1^* = 0$. Hence, (12.21)

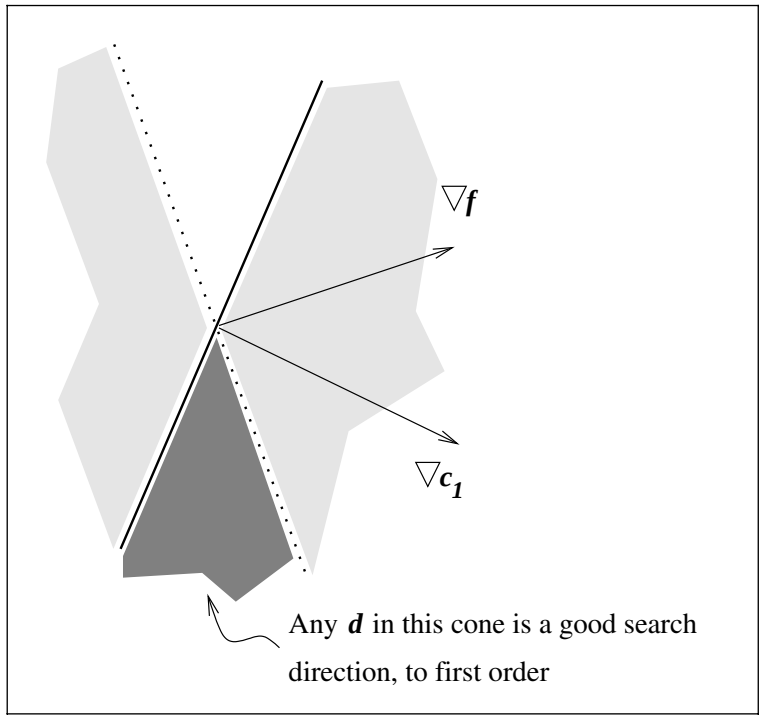


Figure 12.5 A direction d that satisfies both (12.13) and (12.18) lies in the intersection of a closed half-plane and an open half-plane.

reduces to $\nabla f(x^*) = 0$, as required by (12.19). In case II, (12.22) allows λ_1^* to take on a nonnegative value, so (12.21) becomes equivalent to (12.20).

TWO INEQUALITY CONSTRAINTS

❑ EXAMPLE 12.3

Suppose we add an extra constraint to the problem (12.17) to obtain

$$\min\ x_1 + x_2 \qquad \text{s.t.} \qquad 2 - x_1^2 - x_2^2 \geq 0, \quad x_2 \geq 0, \tag{12.23}$$

for which the feasible region is the half-disk illustrated in Figure 12.6. It is easy to see that the solution lies at $(-\sqrt{2}, 0)^T$, a point at which both constraints are active. By repeating the arguments for the previous examples, we conclude that a direction d is a feasible descent

direction, to first order, if it satisfies the following conditions:

$$\nabla c_i(x)^T d \geq 0, \quad i \in \mathcal{I} = \{1, 2\}, \quad \nabla f(x)^T d < 0. \tag{12.24}$$

However, it is clear from Figure 12.6 that no such direction can exist when $x = (-\sqrt{2}, 0)^T$. The conditions $\nabla c_i(x)^T d \geq 0$, $i = 1, 2$, are both satisfied only if d lies in the quadrant defined by $\nabla c_1(x)$ and $\nabla c_2(x)$, but it is clear by inspection that all vectors d in this quadrant satisfy $\nabla f(x)^T d \geq 0$.

Let us see how the Lagrangian and its derivatives behave for the problem (12.23) and the solution point $(-\sqrt{2}, 0)^T$. First, we include an additional term $\lambda_i c_i(x)$ in the Lagrangian for each additional constraint, so we have

$$\mathcal{L}(x, \lambda) = f(x) - \lambda_1 c_1(x) - \lambda_2 c_2(x),$$

where $\lambda = (\lambda_1, \lambda_2)^T$ is the vector of Lagrange multipliers. The extension of condition (12.21) to this case is

$$\nabla_x \mathcal{L}(x^*, \lambda^*) = 0, \quad \text{for some } \lambda^* \geq 0, \tag{12.25}$$

where the inequality $\lambda^* \geq 0$ means that all components of λ^* are required to be nonnegative. By applying the complementarity condition (12.22) to both inequality constraints, we obtain

$$\lambda_1^* c_1(x^*) = 0, \quad \lambda_2^* c_2(x^*) = 0. \tag{12.26}$$

When $x^* = (-\sqrt{2}, 0)^T$, we have

$$\nabla f(x^*) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \nabla c_1(x^*) = \begin{bmatrix} 2\sqrt{2} \\ 0 \end{bmatrix}, \quad \nabla c_2(x^*) = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

so that it is easy to verify that $\nabla_x \mathcal{L}(x^*, \lambda^*) = 0$ when we select λ^* as follows:

$$\lambda^* = \begin{bmatrix} 1/(2\sqrt{2}) \\ 1 \end{bmatrix}.$$

Note that both components of λ^* are positive.

We consider now some other feasible points that are *not* solutions of (12.23), and examine the properties of the Lagrangian and its gradient at these points.

For the point $x = (\sqrt{2}, 0)^T$, we again have that both constraints are active. However, the objective gradient $\nabla f(x)$ no longer lies in the quadrant defined by the conditions $\nabla c_i(x)^T d \geq 0$, $i = 1, 2$ (see Figure 12.7). One first-order feasible descent direction from this point—a vector d that satisfies (12.24)—is simply $d = (-1, 0)^T$; there are many others (see the exercises). For this value of x it is easy to verify that the condition $\nabla_x \mathcal{L}(x, \lambda) = 0$

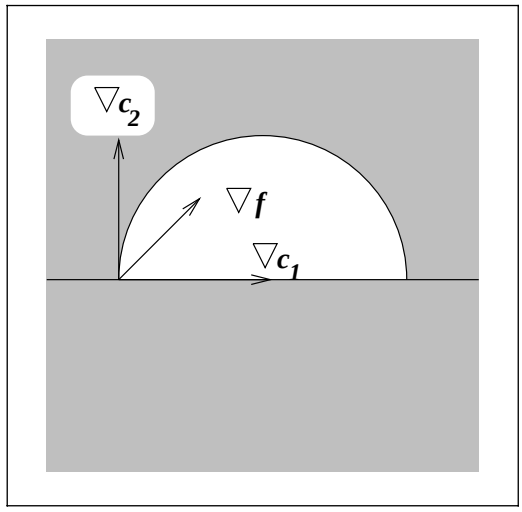


Figure 12.6
Problem (12.23), illustrating the gradients of the active constraints and objective at the solution.

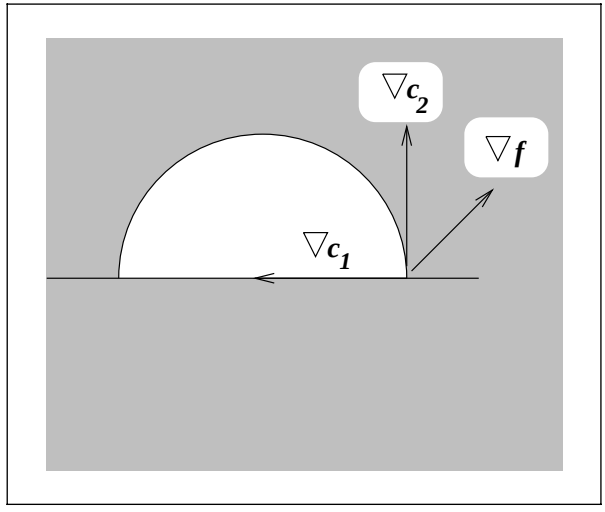


Figure 12.7
Problem (12.23), illustrating the gradients of the active constraints and objective at a nonoptimal point.

is satisfied only when $\lambda = (-1/(2\sqrt{2}), 1)$. Note that the first component λ_1 is negative, so that the conditions (12.25) are not satisfied at this point.

Finally, let us consider the point $x = (1, 0)^T$, at which only the second constraint c_2 is active. At this point, linearization of f and c as in Example 12.2 gives the following conditions, which must be satisfied for d to be a feasible descent direction, to first order:

$$1 + \nabla c_1(x)^T d \geq 0, \quad \nabla c_2(x)^T d \geq 0, \quad \nabla f(x)^T d < 0. \tag{12.27}$$

In fact, we need worry only about satisfying the second and third conditions, since we can always satisfy the first condition by multiplying d by a sufficiently small positive quantity. By noting that

$$\nabla f(x) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \nabla c_2(x) = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

it is easy to verify that the vector $d = \left(-\frac{1}{2}, \frac{1}{4}\right)$ satisfies (12.27) and is therefore a descent direction.

To show that optimality conditions (12.25) and (12.26) fail, we note first from (12.26) that since $c_1(x) > 0$, we must have $\lambda_1 = 0$. Therefore, in trying to satisfy $\nabla_x \mathcal{L}(x, \lambda) = 0$, we are left to search for a value λ_2 such that $\nabla f(x) - \lambda_2 \nabla c_2(x) = 0$. No such λ_2 exists, and thus this point fails to satisfy the optimality conditions. □

12.2 FIRST-ORDER OPTIMALITY CONDITIONS

STATEMENT OF FIRST-ORDER NECESSARY CONDITIONS

The three examples above suggest that a number of conditions are important in the characterization of solutions for (12.1). These include the relation $\nabla_x \mathcal{L}(x, \lambda) = 0$, the nonnegativity of λ_i for all inequality constraints $c_i(x)$, and the complementarity condition $\lambda_i c_i(x) = 0$ that is required for all the inequality constraints. We now generalize the observations made in these examples and state the first-order optimality conditions in a rigorous fashion.

In general, the Lagrangian for the constrained optimization problem (12.1) is defined as

$$\mathcal{L}(x, \lambda) = f(x) - \sum_{i \in \mathcal{E} \cup \mathcal{I}} \lambda_i c_i(x). \tag{12.28}$$

The *active set* $\mathcal{A}(x)$ at any feasible x is the union of the set $\mathcal{E}$ with the indices of the active inequality constraints; that is,

$$\mathcal{A}(x) = \mathcal{E} \cup \{i \in \mathcal{I} \mid c_i(x) = 0\}. \tag{12.29}$$

Next, we need to give more attention to the properties of the constraint gradients. The vector $\nabla c_i(x)$ is often called the *normal* to the constraint c_i at the point x, because it is usually a vector that is perpendicular to the contours of the constraint c_i at x, and in the case of an inequality constraint, it points toward the feasible side of this constraint. It is possible, however, that $\nabla c_i(x)$ vanishes due to the algebraic representation of c_i, so that the term

$\lambda_i \nabla c_i(x)$ vanishes for all values of λ_i and does not play a role in the Lagrangian gradient $\nabla_x \mathcal{L}$. For instance, if we replaced the constraint in (12.9) by the equivalent condition

$$c_1(x) = \left(x_1^2 + x_2^2 - 2\right)^2 = 0,$$

we would have that $\nabla c_1(x) = 0$ for all feasible points x, and in particular that the condition $\nabla f(x) = \lambda_1 \nabla c_1(x)$ no longer holds at the optimal point $(-1, -1)^T$. We usually make an assumption called a *constraint qualification* to ensure that such degenerate behavior does not occur at the value of x in question. One such constraint qualification—probably the one most often used in the design of algorithms—is the one defined as follows:

Definition 12.1 (LICQ).

Given the point x^* and the active set $\mathcal{A}(x^*)$ defined by (12.29), we say that the linear independence constraint qualification (LICQ) *holds if the set of active constraint gradients $\{\nabla c_i(x^*), i \in \mathcal{A}(x^*)\}$ is linearly independent.*

Note that if this condition holds, none of the active constraint gradients can be zero.

This condition allows us to state the following optimality conditions for a general nonlinear programming problem (12.1). These conditions provide the foundation for many of the algorithms described in the remaining chapters of the book. They are called *first-order conditions* because they concern themselves with properties of the gradients (first-derivative vectors) of the objective and constraint functions.

Theorem 12.1 (First-Order Necessary Conditions).

Suppose that x^* is a local solution of (12.1) and that the LICQ holds at x^*. Then there is a Lagrange multiplier vector λ^*, with components λ_i^*, $i \in \mathcal{E} \cup \mathcal{I}$, such that the following conditions are satisfied at (x^*, λ^*)

$$\nabla_x \mathcal{L}(x^*, \lambda^*) = 0, \tag{12.30a}$$
$$c_i(x^*) = 0, \quad \text{for all } i \in \mathcal{E}, \tag{12.30b}$$
$$c_i(x^*) \geq 0, \quad \text{for all } i \in \mathcal{I}, \tag{12.30c}$$
$$\lambda_i^* \geq 0, \quad \text{for all } i \in \mathcal{I}, \tag{12.30d}$$
$$\lambda_i^* c_i(x^*) = 0, \quad \text{for all } i \in \mathcal{E} \cup \mathcal{I}. \tag{12.30e}$$

The conditions (12.30) are often known as the *Karush–Kuhn–Tucker conditions*, or *KKT conditions* for short. Because the complementarity condition implies that the Lagrange multipliers corresponding to inactive inequality constraints are zero, we can omit the terms for indices $i \notin \mathcal{A}(x^*)$ from (12.30a) and rewrite this condition as

$$0 = \nabla_x \mathcal{L}(x^*, \lambda^*) = \nabla f(x^*) - \sum_{i \in \mathcal{A}(x^*)} \lambda_i^* \nabla c_i(x^*). \tag{12.31}$$

A special case of complementarity is important and deserves its own definition:

Definition 12.2 (Strict Complementarity).

Given a local solution x^* of (12.1) and a vector λ^* satisfying (12.30), we say that the strict complementarity condition *holds if* exactly one *of λ_i^* and $c_i(x^*)$ is zero for each index $i \in \mathcal{I}$. In other words, we have that $\lambda_i^* > 0$ for each $i \in \mathcal{I} \cap \mathcal{A}(x^*)$.*

For a given problem (12.1) and solution point x^*, there may be many vectors λ^* for which the conditions (12.30) are satisfied. When the LICQ holds, however, the optimal λ^* is unique (see the exercises).

The proof of Theorem 12.1 is quite complex, but it is important to our understanding of constrained optimization, so we present it in the next section. First, we illustrate the KKT conditions with another example.

❑ EXAMPLE 12.4

Consider the feasible region illustrated in Figure 12.2 and described by the four constraints (12.6). By restating the constraints in the standard form of (12.1) and including an objective function, the problem becomes

$$\min_x \left(x_1 - \frac{3}{2}\right)^2 + \left(x_2 - \frac{1}{8}\right)^4 \qquad \text{s.t.} \qquad \begin{bmatrix} 1 - x_1 - x_2 \\ 1 - x_1 + x_2 \\ 1 + x_1 - x_2 \\ 1 + x_1 + x_2 \end{bmatrix} \geq 0. \tag{12.32}$$

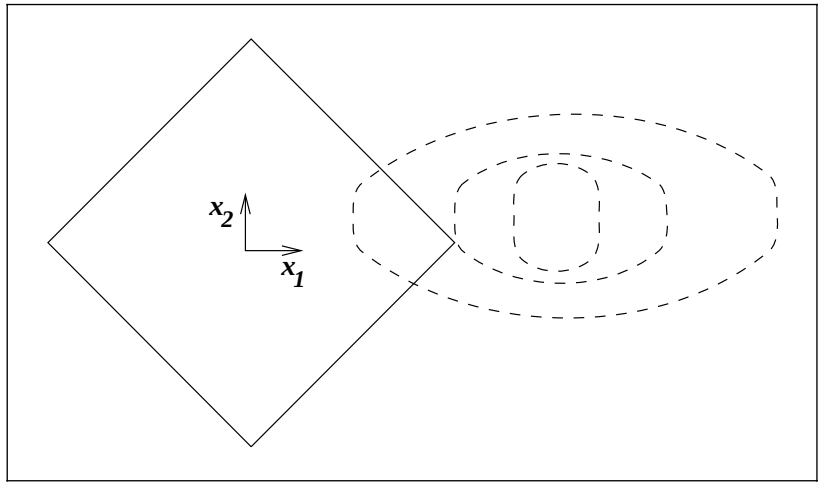


Figure 12.8 Inequality-constrained problem (12.32) with solution at $(1, 0)^T$.

It is fairly clear from Figure 12.8 that the solution is $x^* = (1, 0)$. The first and second constraints in (12.32) are active at this point. Denoting them by c_1 and c_2 (and the inactive constraints by c_3 and c_4), we have

$$\nabla f(x^*) = \begin{bmatrix} -1 \\ -\frac{1}{2} \end{bmatrix}, \quad \nabla c_1(x^*) = \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \quad \nabla c_2(x^*) = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

Therefore, the KKT conditions (12.30a)-(12.30e) are satisfied when we set

$$\lambda^* = \left(\tfrac{3}{4}, \tfrac{1}{4}, 0, 0\right)^T.$$

□

SENSITIVITY

The convenience of using Lagrange multipliers should now be clear, but what of their intuitive significance? The value of each Lagrange multiplier λ_i^* tells us something about the *sensitivity* of the optimal objective value $f(x^*)$ to the presence of constraint c_i. To put it another way, λ_i^* indicates how hard f is "pushing" or "pulling" against the particular constraint c_i. We illustrate this point with a little analysis. When we choose an inactive constraint $i \notin \mathcal{A}(x^*)$ such that $c_i(x^*) > 0$, the solution x^* and function value $f(x^*)$ are quite indifferent to whether this constraint is present or not. If we perturb c_i by a tiny amount, it will still be inactive and x^* will still be a local solution of the optimization problem. Since $\lambda_i^* = 0$ from (12.30e), the Lagrange multiplier indicates accurately that constraint i is not significant.

Suppose instead that constraint i is active, and let us perturb the right-hand-side of this constraint a little, requiring, say, that $c_i(x) \geq -\epsilon \|\nabla c_i(x^*)\|$ instead of $c_i(x) \geq 0$. Suppose that ϵ is sufficiently small that the perturbed solution $x^*(\epsilon)$ still has the same set of active constraints, and that the Lagrange multipliers are not much affected by the perturbation. (These conditions can be made more rigorous with the help of strict complementarity and second-order conditions, as discussed later in the chapter.) We then find that

$$\begin{aligned} -\epsilon \|\nabla c_i(x^*)\| = c_i(x^*(\epsilon)) - c_i(x^*) &\approx (x^*(\epsilon) - x^*)^T \nabla c_i(x^*), \\ 0 = c_j(x^*(\epsilon)) - c_j(x^*) &\approx (x^*(\epsilon) - x^*)^T \nabla c_j(x^*), \\ &\text{for all } j \in \mathcal{A}(x^*) \text{ with } j \neq i. \end{aligned}$$

The value of $f(x^*(\epsilon))$, meanwhile, can be estimated with the help of (12.30a). We have

$$\begin{aligned} f(x^*(\epsilon)) - f(x^*) &\approx (x^*(\epsilon) - x^*)^T \nabla f(x^*) \\ &= \sum_{j \in \mathcal{A}(x^*)} \lambda_j^* (x^*(\epsilon) - x^*)^T \nabla c_j(x^*) \\ &\approx -\epsilon \|\nabla c_i(x^*)\| \lambda_i^*. \end{aligned}$$

By taking limits, we see that the family of solutions $x^*(\epsilon)$ satisfies

$$\frac{df(x^*(\epsilon))}{d\epsilon} = -\lambda_i^* \|\nabla c_i(x^*)\|. \tag{12.33}$$

A sensitivity analysis of this problem would conclude that if $\lambda_i^* \|\nabla c_i(x^*)\|$ is large, then the optimal value is sensitive to the placement of the ith constraint, while if this quantity is small, the dependence is not too strong. If λ_i^* is exactly zero for some active constraint, small perturbations to c_i in some directions will hardly affect the optimal objective value at all; the change is zero, to first order.

This discussion motivates the definition below, which classifies constraints according to whether or not their corresponding Lagrange multiplier is zero.

Definition 12.3.

Let x^* be a solution of the problem (12.1), and suppose that the KKT conditions (12.30) are satisfied. We say that an inequality constraint c_i is strongly active *or* binding *if $i \in \mathcal{A}(x^*)$ and $\lambda_i^* > 0$ for some Lagrange multiplier λ^* satisfying (12.30). We say that c_i is* weakly active *if $i \in \mathcal{A}(x^*)$ and $\lambda_i^* = 0$ for all λ^* satisfying (12.30).*

Note that the analysis above is independent of scaling of the individual constraints. For instance, we might change the formulation of the problem by replacing some active constraint c_i by $10c_i$. The new problem will actually be equivalent (that is, it has the same feasible set and same solution), but the optimal multiplier λ_i^* corresponding to c_i will be replaced by $\lambda_i^*/10$. However, since $\|\nabla c_i(x^*)\|$ is replaced by $10\|\nabla c_i(x^*)\|$, the product $\lambda_i^* \|\nabla c_i(x^*)\|$ does not change. If, on the other hand, we replace the objective function f by $10f$, the multipliers λ_i^* in (12.30) all will need to be replaced by $10\lambda_i^*$. Hence in (12.33) we see that the sensitivity of f to perturbations has increased by a factor of 10, which is exactly what we would expect.

12.3 DERIVATION OF THE FIRST-ORDER CONDITIONS

Having studied some motivating examples, observed the characteristics of optimal and nonoptimal points, and stated the KKT conditions, we now describe a complete proof of Theorem 12.1. This analysis is not just of esoteric interest, but is rather the key to understanding all constrained optimization algorithms.

FEASIBLE SEQUENCES

The first concept we introduce is that of a *feasible sequence*. Given a feasible point x^*, a sequence $\{z_k\}_{k=0}^{\infty}$ with $z_k \in \mathbb{R}^n$ is a feasible sequence if the following properties hold:

(i) $z_k \neq x^*$ for all k;

(ii) $\lim_{k\to\infty} z_k = x^*$;

(iii) z_k is feasible for all sufficiently large values of k.

For later reference, we denote the set of all possible feasible sequences approaching x by $\mathcal{T}(x)$.

We characterize a local solution of (12.1) as *a point x at which all feasible sequences have the property that $f(z_k) \geq f(x)$ for all k sufficiently large.* We derive practical, verifiable conditions under which this property holds. To do so we will make use of the concept of a *limiting direction* of a feasible sequence.

Limiting directions of a feasible sequence are vectors d such that we have

$$\lim_{z_k \in \mathcal{S}_d} \frac{z_k - x}{\|z_k - x\|} \rightarrow d, \tag{12.34}$$

where $\mathcal{S}_d$ is some subsequence of $\{z_k\}_{k=0}^{\infty}$. In general, a feasible sequence has at least one limiting direction and may have more than one. To see this, note that the sequence of vectors defined by

$$d_k = \frac{z_k - x}{\|z_k - x\|}$$

lies on the surface of the unit sphere, which is a compact set, and thus there is at least one limit point d. Moreover, all such points are limiting directions by the definition (12.34). If we have some sequence $\{z_k\}$ with limiting direction d and corresponding subsequence $\mathcal{S}_d$, we can construct another feasible sequence $\{\bar{z}_k\}$ such that

$$\lim_{k \rightarrow \infty} \frac{\bar{z}_k - x}{\|\bar{z}_k - x\|} = d$$

(that is, with a unique limit point) by simply defining each $\bar{z}_k$ to be an element from the subsequence $\mathcal{S}_d$.

We illustrate these concepts by revisiting Example 12.1

❑ EXAMPLE 12.5 (EXAMPLE 12.1, REVISITED)

Figure 12.9 shows a closeup of the problem (12.9), the equality-constrained problem in which the feasible set is a circle of radius $\sqrt{2}$, near the nonoptimal point $x = (-\sqrt{2}, 0)^T$. The figure also shows a feasible sequence approaching x. This sequence could be defined analytically by the formula

$$z_k = \begin{bmatrix} -\sqrt{2 - 1/k^2} \\ -1/k \end{bmatrix}. \tag{12.35}$$

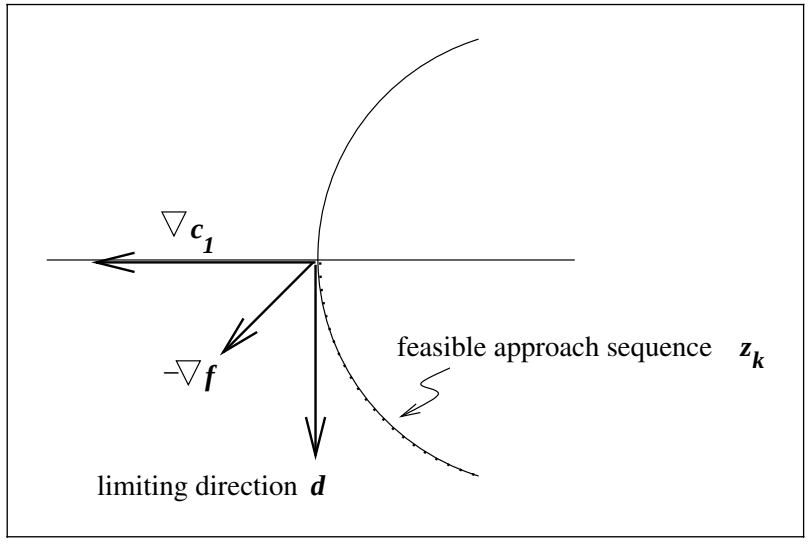


Figure 12.9 Constraint normal, objective gradient, and feasible sequence for problem (12.9).

The vector $d = (0, -1)$ plotted in Figure 12.9 is a limiting direction of this feasible sequence. Note that d is tangent to the feasible sequence at x but points in the opposite direction. The objective function $f(x) = x_1 + x_2$ increases as we move along the sequence (12.35); in fact, we have $f(z_{k+1}) > f(z_k)$ for all $k = 2, 3, \ldots$ (see the exercises). It follows that $f(z_k) < f(x)$ for $k = 2, 3, \ldots$. Hence, x cannot be a solution of (12.9).

Another feasible sequence is one that approaches $x^* = (-\sqrt{2}, 0)$ from the opposite direction. Its elements are defined by

$$z_k = \begin{bmatrix} -\sqrt{2 - 1/k^2} \\ 1/k \end{bmatrix}.$$

It is easy to show that f *decreases* along this sequence and that its limiting direction is $d = (0, 1)^T$. Other feasible sequences are obtained by combining elements from the two sequences already discussed, for instance

$$z_k = \begin{cases} (-\sqrt{2 - 1/k^2}, 1/k), & \text{when } k \text{ is a multiple of 3,} \\ (-\sqrt{2 - 1/k^2}, -1/k), & \text{otherwise.} \end{cases}$$

In general, feasible sequences of points approaching $(-\sqrt{2}, 0)^T$ will have two limiting directions, $(0, 1)^T$ and $(0, -1)^T$.



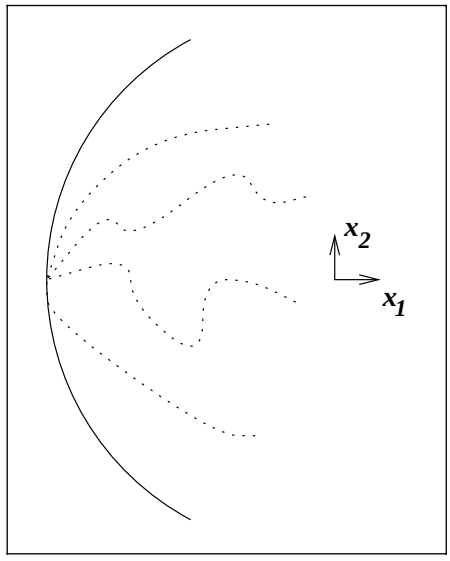


Figure 12.10
Feasible sequences converging to a particular feasible point for the region defined by $x_1^2 + x_2^2 \leq 2$.

We now consider feasible sequences and limiting directions for an example that involves inequality constraints.

❑ EXAMPLE 12.6 (EXAMPLE 12.2, REVISITED)

We now reconsider problem (12.17) in Example 12.2. The solution $x^* = (-1, -1)^T$ is the same as in the equality-constrained case, but there is a much more extensive collection of feasible sequences that converge to any given feasible point (see Figure 12.10). From the point $x = (-\sqrt{2}, 0)^T$, the various feasible sequences defined above for the equality-constrained problem are still feasible for (12.17). There are also infinitely many feasible sequences that converge to $x = (-\sqrt{2}, 0)^T$ along a straight line from the interior of the circle. These are defined by

$$z_k = (-1, 0)^T + (1/k)w, \tag{12.36}$$

where w is any vector whose first component is positive ($w_1 > 0$). Now, z_k is feasible in (12.36), provided that $\|z_k\| \leq 1$, that is,

$$(-1 + w_1/k)^2 + (w_2/k)^2 \leq 1,$$

a condition that is satisfied, provided that $k > (2w_1)/(w_1^2 + w_2^2)$. In addition to these straight-line feasible sequences, we can also define an infinite variety of sequences that

approach $(-\sqrt{2}, 0)$ along a curve from the interior of the circle or that make the approach in a seemingly random fashion.

Given a point x, if it is possible to choose a feasible sequence from $\mathcal{T}(x)$ such that the first-order approximation to the objective function actually *increases* monotonically along the sequence, then x must not be optimal. This condition is the fundamental first-order necessary condition, and we state it formally in the following theorem.

Theorem 12.2.

If x^* is a local solution of (12.1), then all feasible sequences $\{z_k\}$ in $\mathcal{T}(x^*)$ must satisfy

$$\nabla f(x^*)^T d \geq 0, \tag{12.37}$$

where d is any limiting direction of the feasible sequence.

PROOF. Suppose that there is a feasible sequence $\{z_k\}$ with the property $\nabla f(x^*)^T d < 0$, for some limiting direction d, and let $\mathcal{S}_d$ be the subsequence of $\{z_k\}$ that approaches x^*. By Taylor's theorem (Theorem 2.1), we have for any $z_k \in \mathcal{S}_d$ that

$$\begin{aligned} f(z_k) &= f(x^*) + (z_k - x^*)^T \nabla f(x^*) + o(\|z_k - x^*\|) \\ &= f(x^*) + \|z_k - x^*\| d^T \nabla f(x^*) + o(\|z_k - x^*\|). \end{aligned}$$

Since $d^T \nabla f(x^*) < 0$, we have that the remainder term is eventually dominated by the first-order term, that is,

$$f(z_k) < f(x^*) + \tfrac{1}{2}\|z_k - x^*\| d^T \nabla f(x^*), \qquad \text{for all } k \text{ sufficiently large.}$$

Hence, given any open neighborhood of x^*, we can choose k sufficiently large that z_k lies within this neighborhood and has a lower value of the objective f. Therefore, x^* is not a local solution. □

This theorem tells us why we can ignore constraints that are *strictly inactive* (that is, constraints for which $c_i(x) > 0$) in formulating optimality conditions. The theorem does not use the whole range of properties of the feasible sequence, but rather one specific property: the limiting directions of $\{z_k\}$. Because of the way in which the limiting directions are defined, it is clear that only the asymptotic behavior of the sequence is relevant, that is, its behavior for large values of the index k. If some constraint $i \in \mathcal{I}$ is inactive at x, then we have $c_i(z_k) > 0$ for all k sufficiently large, so that a constraint that is inactive at x is also inactive at all sufficiently advanced elements of the feasible sequence $\{z_k\}$.

CHARACTERIZING LIMITING DIRECTIONS: CONSTRAINT QUALIFICATIONS

Theorem 12.2 is quite general, but it is not very useful as stated, because it seems to require knowledge of all possible limiting directions for all feasible sequences $\mathcal{T}(x^*)$. In this section we show that *constraint qualifications* allow us to characterize the salient properties of $\mathcal{T}(x^*)$, and therefore make the condition (12.37) easier to verify.

One frequently used constraint qualification is the *linear independence constrained qualification* (LICQ) given in Definition 12.1. The following lemma shows that when LICQ holds, there is a neat way to characterize the set of all possible limiting directions d in terms of the gradients $\nabla c_i(x^*)$ of the active constraints at x^*.

In subsequent results we introduce the notation A to represent the matrix whose rows are the active constraint gradients at the optimal point, that is,

$$\nabla c_i^* = \nabla c_i(x^*), \quad A^T = [\nabla c_i^*]_{i \in \mathcal{A}(x^*)}, \quad \nabla f^* = \nabla f(x^*), \tag{12.38}$$

where the active set $\mathcal{A}(x^*)$ is defined as in (12.29).

Lemma 12.3.

The following two statements are true.

(i) *If $d \in \mathbb{R}^n$ is a limiting direction of a feasible sequence, then*

$$d^T \nabla c_i^* = 0, \quad \text{for all } i \in \mathcal{E}, \quad d^T \nabla c_i^* \geq 0, \quad \text{for all } i \in \mathcal{A}(x^*) \cap \mathcal{I}. \tag{12.39}$$

(ii) *If (12.39) holds with $\|d\| = 1$ and the LICQ condition is satisfied, then $d \in \mathbb{R}^n$ is a limiting direction of some feasible sequence.*

PROOF. Without loss of generality, let us assume that all the constraints $c_i(\cdot)$, $i = 1, 2, \ldots, m$, are active. (We can arrive at this convenient ordering by simply dropping all inactive constraints—which are irrelevant in some neighborhood of x^*—and renumbering the active constraints that remain.)

To prove (i), let $\{z_k\} \in \mathcal{T}(x^*)$ be some feasible sequence for which d is a limiting direction, and assume (by taking a subsequence if necessary) that

$$\lim_{k \to \infty} \frac{z_k - x^*}{\|z_k - x^*\|} = d.$$

From this definition, we have that

$$z_k = x^* + \|z_k - x^*\| d + o(\|z_k - x^*\|).$$

By taking $i \in \mathcal{E}$ and using Taylor's theorem, we have that

$$\begin{aligned} 0 &= \frac{1}{\|z_k - x^*\|} c_i(z_k) \\ &= \frac{1}{\|z_k - x^*\|} \left[c_i(x^*) + \|z_k - x^*\| \nabla c_i^T d + o(\|z_k - x^*\|) \right] \\ &= \nabla c_i^T d + \frac{o(\|z_k - x^*\|)}{\|z_k - x^*\|}. \end{aligned}$$

By taking the limit as $k \to \infty$, the last term in this expression vanishes, and we have $\nabla c_i^T d = 0$, as required. For the active inequality constraints $i \in \mathcal{A}(x^*) \cap \mathcal{I}$, we have similarly that

$$\begin{aligned} 0 &\leq \frac{1}{\|z_k - x^*\|} c_i(z_k) \\ &= \frac{1}{\|z_k - x^*\|} \left[c_i(x^*) + \|z_k - x^*\| \nabla c_i^T d + o(\|z_k - x^*\|) \right] \\ &= \nabla c_i^T d + \frac{o(\|z_k - x^*\|)}{\|z_k - x^*\|}. \end{aligned}$$

Hence, by a similar limiting argument, we have that $\nabla c_i^T d \geq 0$, as required.

For (ii), we use the implicit function theorem (see the Appendix or Lang [147, p. 131] for a statement of this result). First, since the LICQ holds, we have from Definition 12.1 that the $m \times n$ matrix A of active constraint gradients has full row rank m. Let Z be a matrix whose columns are a basis for the null space of A; that is,

$$Z \in \mathbb{R}^{n \times (n-m)}, \qquad Z \text{ has full column rank}, \qquad AZ = 0. \tag{12.40}$$

Let d have the properties (12.39), and suppose that $\{t_k\}_{k=0}^{\infty}$ is any sequence of positive scalars such $\lim_{k \to \infty} t_k = 0$. Define the parametrized system of equations $R : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^n$ by

$$R(z, t) = \begin{bmatrix} c(z) - tAd \\ Z^T(z - x^* - td) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \tag{12.41}$$

We claim that for each $t = t_k$, the solutions $z = z_k$ of this system for small $t > 0$ give a feasible sequence that approaches x^*.

Clearly, for $t = 0$, the solution of (12.41) is $z = x^*$, and the Jacobian of R at this point is

$$\nabla_z R(x^*, 0) = \begin{bmatrix} A \\ Z^T \end{bmatrix}, \tag{12.42}$$

which is nonsingular by construction of Z. Hence, according to the implicit function theorem, the system (12.41) has a unique solution z_k for all values of t_k sufficiently small. Moreover, we have from (12.41) and (12.39) that

$$i \in \mathcal{E} \Rightarrow c_i(z_k) = t_k \nabla c_i^T d = 0, \tag{12.43a}$$

$$i \in \mathcal{A}(x^*) \cap \mathcal{I} \Rightarrow c_i(z_k) = t_k \nabla c_i^T d \geq 0, \tag{12.43b}$$

so that z_k is indeed feasible. Also, for any positive value $t = \bar{t} > 0$, we cannot have $z(t) = x^*$, since otherwise by substituting $(z, t) = (x^*, \bar{t})$ into (12.41), we obtain

$$\begin{bmatrix} c(x^*) - \bar{t} A d \\ -Z^T(\bar{t} d) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Since $c(x^*) = 0$ (we have assumed that all constraints are active) and $\bar{t} > 0$, we have from the full rank of the matrix in (12.42) that $d = 0$, which contradicts $\|d\| = 1$. It follows that $z_k = z(t_k) \neq x^*$ for all k.

It remains to show that d is a limiting direction of $\{z_k\}$. Using the fact that $R(z_k, t_k) = 0$ for all k together with Taylor's theorem, we find that

$$\begin{aligned} 0 = R(z_k, t_k) &= \begin{bmatrix} c(z_k) - t_k A d \\ Z^T(z_k - x^* - t_k d) \end{bmatrix} \\ &= \begin{bmatrix} A(z_k - x^*) + o(\|z_k - x^*\|) - t_k A d \\ Z^T(z_k - x^* - t_k d) \end{bmatrix} \\ &= \begin{bmatrix} A \\ Z^T \end{bmatrix} (z_k - x^* - t_k d) + o(\|z_k - x^*\|). \end{aligned}$$

By dividing this expression by $\|z_k - x^*\|$ and using nonsingularity of the coefficient matrix in the first term, we obtain

$$\lim_{k \to \infty} d_k - \frac{t_k}{\|z_k - x^*\|} d = 0, \quad \text{where } d_k = \frac{z_k - x^*}{\|z_k - x^*\|}. \tag{12.44}$$

Since $\|d_k\| = 1$ for all k and since $\|d\| = 1$, we must have

$$\lim_{k \to \infty} \frac{t_k}{\|z_k - x^*\|} = 1. \tag{12.45}$$

(We leave the simple proof by contradiction of this statement as an exercise.) Hence, from (12.44), we have $\lim_{k \to \infty} d_k = d$, as required. □

The set of directions defined by (12.39) plays a central role in the optimality conditions, so for future reference we give this set a name and define it formally.

Definition 12.4.

Given a point x^* and the active constraint set $\mathcal{A}(x^*)$ defined by (12.29), the set F_1 is defined by

$$F_1 = \left\{ \alpha d \,\middle|\, \alpha > 0, \quad \begin{array}{ll} d^T \nabla c_i^* = 0, & \text{for all } i \in \mathcal{E}, \\ d^T \nabla c_i^* \geq 0, & \text{for all } i \in \mathcal{A}(x^*) \cap \mathcal{I} \end{array} \right\}.$$

Note that F_1 is a cone. In fact, when a constraint qualification is satisfied, F_1 is the tangent cone to the feasible set at x^*. (See Section 12.6 and the definition (A.1) in the Appendix.)

INTRODUCING LAGRANGE MULTIPLIERS

Lemma 12.3 tells us that when the LICQ holds, the cone F_1 is simply the set of all positive multiples of all limiting directions of all possible feasible sequences. Therefore, the condition (12.37) of Theorem 12.2 holds if $\nabla f(x^*)^T d < 0$ for all $d \in F_1$. This condition, too, would appear to be impossible to check, since the set F_1 contains infinitely many vectors in general. The next lemma gives an alternative, practical way to check this condition that makes use of the Lagrange multipliers, the variables λ_i that were introduced in the definition (12.28) of the Lagrangian $\mathcal{L}$.

Lemma 12.4.

There is no direction $d \in F_1$ for which $d^T \nabla f^* < 0$ if and only if there exists a vector $\lambda \in \mathbb{R}^m$ with

$$\nabla f^* = \sum_{i \in \mathcal{A}(x^*)} \lambda_i \nabla c_i^* = A(x^*)^T \lambda, \qquad \lambda_i \geq 0 \ \text{ for } i \in \mathcal{A}(x^*) \cap \mathcal{I}. \tag{12.46}$$

PROOF. If we define the cone N by

$$N = \left\{ s \,\middle|\, s = \sum_{i \in \mathcal{A}(x^*)} \lambda_i \nabla c_i^*, \quad \lambda_i \geq 0 \text{ for } i \in \mathcal{A}(x^*) \cap \mathcal{I} \right\}, \tag{12.47}$$

then the condition (12.46) is equivalent to $\nabla f^* \in N$. We note first that the set N is closed—a fact that is intuitively clear but nontrivial to prove rigorously. (An intriguing proof of this claim appears in the Notes and References at the end of this chapter.)

We prove the forward implication by supposing that (12.46) holds and choosing d to be any vector satisfying (12.39). We then have that

$$d^T \nabla f^* = \sum_{i \in \mathcal{E}} \lambda_i (d^T \nabla c_i^*) + \sum_{i \in \mathcal{A}(x^*) \cap \mathcal{I}} \lambda_i (d^T \nabla c_i^*).$$

The first summation is zero because $d^T \nabla c_i^* = 0$ for $i \in \mathcal{E}$, while the second term is nonnegative because $\lambda_i \geq 0$ and $d^T \nabla c_i^* \geq 0$ for $i \in \mathcal{A}(x^*) \cap \mathcal{I}$. Hence $d^T \nabla f^* \geq 0$.

For the reverse implication, we show that if ∇f^* does *not* satisfy (12.46) (that is, $\nabla f^* \notin N$), then we can find a vector d for which $d^T \nabla f^* < 0$ and (12.39) holds.

Let $\hat{s}$ be the vector in N that is closest to ∇f^*. Because N is closed, $\hat{s}$ is well-defined. In fact, $\hat{s}$ solves the constrained optimization problem

$$\min \|s - \nabla f^*\|_2^2 \quad \text{subject to } s \in N. \tag{12.48}$$

Since $\hat{s} \in N$, we also have $t\hat{s} \in N$ for all scalars $t \geq 0$. Since $\|t\hat{s} - \nabla f^*\|_2^2$ is minimized at $t = 1$, we have

$$\begin{aligned} \frac{d}{dt}\|t\hat{s} - \nabla f^*\|_2^2\Big|_{t=1} = 0 \Rightarrow \left(-2\hat{s}^T \nabla f^* + 2t\hat{s}^T\hat{s}\right)\Big|_{t=1} = 0 \\ \Rightarrow \hat{s}^T(\hat{s} - \nabla f^*) = 0. \end{aligned} \tag{12.49}$$

Now, let s be any other vector in N. Since N is convex (check!), we have by the minimizing property of $\hat{s}$ that

$$\|\hat{s} + \theta(s - \hat{s}) - \nabla f^*\|_2^2 \geq \|\hat{s} - \nabla f^*\|_2^2 \quad \text{for all } \theta \in [0, 1],$$

and hence

$$2\theta(s - \hat{s})^T(\hat{s} - \nabla f^*) + \theta^2\|s - \hat{s}\|_2^2 \geq 0.$$

By dividing this expression by θ and taking the limit as $\theta \downarrow 0$, we have $(s - \hat{s})^T(\hat{s} - \nabla f^*) \geq 0$. Therefore, because of (12.49),

$$s^T(\hat{s} - \nabla f^*) \geq 0, \quad \text{for all } s \in N. \tag{12.50}$$

We claim now that the vector

$$d = \hat{s} - \nabla f^*$$

satisfies both (12.39) and $d^T \nabla f^* < 0$. Note that $d \neq 0$ because ∇f^* does not belong to the cone N. We have from (12.49) that

$$d^T \nabla f^* = d^T(\hat{s} - d) = (\hat{s} - \nabla f^*)^T\hat{s} - d^T d = -\|d\|_2^2 < 0,$$

so that d satisfies the descent property.

By making appropriate choices of coefficients λ_i, $i = 1, 2, \ldots, m$, it is easy to see that

$$\begin{aligned} i \in \mathcal{E} \Rightarrow \nabla c_i^* \in N \quad \text{and} \quad -\nabla c_i^* \in N; \\ i \in \mathcal{A}(x^*) \cap \mathcal{I} \Rightarrow \nabla c_i^* \in N. \end{aligned}$$

Hence, from (12.50), we have by substituting $d = \hat{s} - \nabla f^*$ and the particular choices $s = \nabla c_i^*$ and $s = -\nabla c_i^*$ that

$$\begin{aligned} i \in \mathcal{E} &\Rightarrow d^T \nabla c_i^* \geq 0 \quad \text{and} \quad -d^T \nabla c_i^* \geq 0 \Rightarrow d^T \nabla c_i^* = 0; \\ i \in \mathcal{A}(x^*) \cap \mathcal{I} &\Rightarrow d^T \nabla c_i^* \geq 0. \end{aligned}$$

Therefore, d also satisfies (12.39), so the reverse implication is proved. □

PROOF OF THEOREM 12.1

Lemmas 12.3 and 12.4 can be combined to give the KKT conditions described in Theorem 12.1. Suppose that $x^* \in \mathbb{R}^n$ is a feasible point at which the LICQ holds. The theorem claims that if x^* is a local solution for (12.1), then there is a vector $\lambda^* \in \mathbb{R}^m$ that satisfies the conditions (12.30).

We show first that there are multipliers λ_i, $i \in \mathcal{A}(x^*)$, such that (12.46) is satisfied. Theorem 12.2 tells us that $d^T \nabla f^* \geq 0$ for all vectors d that are limiting directions of feasible sequences. From Lemma 12.3, we know that when LICQ holds, the set of all possible limiting directions is exactly the set of vectors that satisfy the conditions (12.39). By putting these two statements together, we find that all directions d that satisfy (12.39) must also have $d^T \nabla f^* \geq 0$. Hence, from Lemma 12.4, we have that there is a vector λ for which (12.46) holds, as claimed.

We now define the vector λ^* by

$$\lambda_i^* = \begin{cases} \lambda_i, & i \in \mathcal{A}(x^*), \\ 0, & \text{otherwise}, \end{cases} \tag{12.51}$$

and show that this choice of λ^*, together with our local solution x^*, satisfies the conditions (12.30). We check these conditions in turn.

- The condition (12.30a) follows immediately from (12.46) and the definitions (12.28) of the Lagrangian function and (12.51) of λ^*.
- Since x^* is feasible, the conditions (12.30b) and (12.30c) are satisfied.
- We have from (12.46) that $\lambda_i^* \geq 0$ for $i \in \mathcal{A}(x^*) \cap \mathcal{I}$, while from (12.51), $\lambda_i^* = 0$ for $i \in \mathcal{I} \backslash \mathcal{A}(x^*)$. Hence, $\lambda_i^* \geq 0$ for $i \in \mathcal{I}$, so that (12.30d) holds.
- We have for $i \in \mathcal{A}(x^*) \cap \mathcal{I}$ that $c_i(x^*) = 0$, while for $i \in \mathcal{I} \backslash \mathcal{A}(x^*)$, we have $\lambda_i^* = 0$. Hence $\lambda_i^* c_i(x^*) = 0$ for $i \in \mathcal{I}$, so that (12.30e) is satisfied as well.

This completes the proof.

12.4 SECOND-ORDER CONDITIONS

So far, we have described the first-order conditions—the KKT conditions—which tell us how the first derivatives of f and the active constraints c_i are related at x^*. When these conditions are satisfied, a move along any vector w from F_1 either increases the first-order approximation to the objective function (that is, $w^T \nabla f(x^*) > 0$), or else keeps this value the same (that is, $w^T \nabla f(x^*) = 0$).

What implications does optimality have for the *second* derivatives of f and the constraints c_i? We see in this section that these derivatives play a "tiebreaking" role. For the directions $w \in F_1$ for which $w^T \nabla f(x^*) = 0$, we cannot determine from first derivative information alone whether a move along this direction will increase or decrease the objective function f. Second-order conditions examine the second derivative terms in the Taylor series expansions of f and c_i, to see whether this extra information resolves the issue of increase or decrease in f. Essentially, the second-order conditions concern the curvature of the Lagrangian function in the "undecided" directions—the directions $w \in F_1$ for which $w^T \nabla f(x^*) = 0$.

Since we are discussing second derivatives, stronger smoothness assumptions are needed here than in the previous sections. For the purpose of this section, f and $c_i, i \in \mathcal{E} \cup \mathcal{I}$, are all assumed to be twice continuously differentiable.

Given F_1 from Definition 12.4 and some Lagrange multiplier vector λ^* satisfying the KKT conditions (12.30), we define a subset $F_2(\lambda^*)$ of F_1 by

$$F_2(\lambda^*) = \{w \in F_1 \mid \nabla c_i(x^*)^T w = 0, \text{ all } i \in \mathcal{A}(x^*) \cap \mathcal{I} \text{ with } \lambda_i^* > 0\}.$$

Equivalently,

$$w \in F_2(\lambda^*) \;\Leftrightarrow\; \begin{cases} \nabla c_i(x^*)^T w = 0, & \text{for all } i \in \mathcal{E}, \\ \nabla c_i(x^*)^T w = 0, & \text{for all } i \in \mathcal{A}(x^*) \cap \mathcal{I} \text{ with } \lambda_i^* > 0, \\ \nabla c_i(x^*)^T w \geq 0, & \text{for all } i \in \mathcal{A}(x^*) \cap \mathcal{I} \text{ with } \lambda_i^* = 0. \end{cases} \tag{12.52}$$

The subset $F_2(\lambda^*)$ contains the directions w that tend to "adhere" to the active inequality constraints for which the Lagrange multiplier component λ_i^* is positive, as well as to the equality constraints. From the definition (12.52) and the fact that $\lambda_i^* = 0$ for all inactive components $i \in \mathcal{I} \backslash \mathcal{A}(x^*)$, it follows immediately that

$$w \in F_2(\lambda^*) \;\Rightarrow\; \lambda_i^* \nabla c_i(x^*)^T w = 0 \;\text{ for all } i \in \mathcal{E} \cup \mathcal{I}. \tag{12.53}$$

Hence, from the first KKT condition (12.30a) and the definition (12.28) of the Lagrangian function, we have that

$$w \in F_2(\lambda^*) \;\Rightarrow\; w^T \nabla f(x^*) = \sum_{i \in \mathcal{E} \cup \mathcal{I}} \lambda_i^* w^T \nabla c_i(x^*) = 0. \tag{12.54}$$

Hence the set $F_2(\lambda^*)$ contains directions from F_1 for which it is not clear from first derivative information alone whether f will increase or decrease.

The first theorem defines a *necessary* condition involving the second derivatives: If x^* is a local solution, then the curvature of the Lagrangian along directions in $F_2(\lambda^*)$ must be nonnegative.

Theorem 12.5 (Second-Order Necessary Conditions).

Suppose that x^* is a local solution of (12.1) and that the LICQ condition is satisfied. Let λ^* be a Lagrange multiplier vector such that the KKT conditions (12.30) are satisfied, and let $F_2(\lambda^*)$ be defined as above. Then

$$w^T \nabla_{xx} \mathcal{L}(x^*, \lambda^*) w \geq 0, \quad \text{for all } w \in F_2(\lambda^*). \tag{12.55}$$

PROOF. Since x^* is a local solution, all feasible sequences $\{z_k\}$ approaching x^* must have $f(z_k) \geq f(x^*)$ for all k sufficiently large. Our approach in this proof is to construct a feasible sequence whose limiting direction is $w/\|w\|$ and show that the property $f(z_k) \geq f(x^*)$ implies that (12.55) holds.

Since $w \in F_2(\lambda^*) \subset F_1$, we can use the technique in the proof of Lemma 12.3 to construct a feasible sequence $\{z_k\}$ such that

$$\lim_{k \to \infty} \frac{z_k - x^*}{\|z_k - x^*\|} = \frac{w}{\|w\|}.$$

In particular, we have from formula (12.43) that

$$c_i(z_k) = \frac{t_k}{\|w\|} \nabla c_i(x^*)^T w, \quad \text{for all } i \in \mathcal{A}(x^*), \tag{12.56}$$

where $\{t_k\}$ is some sequence of positive scalars decreasing to zero. Moreover, we have from (12.45) that

$$\|z_k - x^*\| = t_k + o(t_k) \tag{12.57}$$

(this is an alternative way of stating the limit $\lim_{k \to \infty} t_k / \|z_k - x^*\| = 1$), and so by substitution into (12.44), we obtain

$$z_k - x^* = \frac{t_k}{\|w\|} w + o(t_k). \tag{12.58}$$

From (12.28), (12.30), and (12.56), we have as in (12.31) that

$$\mathcal{L}(z_k, \lambda^*) = f(z_k) - \sum_{i \in \mathcal{E} \cup \mathcal{I}} \lambda_i^* c_i(z_k)$$

$$= f(z_k) - \frac{t_k}{\|w\|} \sum_{i \in \mathcal{A}(x^*)} \lambda_i^* \nabla c_i(x^*)^T w$$
$$= f(z_k), \tag{12.59}$$

where the last equality follows from the critical property (12.53). On the other hand, we can perform a Taylor series expansion to obtain an estimate of $\mathcal{L}(z_k, \lambda^*)$ near x^*. By using the Taylor's theorem expression (2.6) and continuity of the Hessians $\nabla^2 f$ and $\nabla^2 c_i$, $i \in \mathcal{E} \cup \mathcal{I}$, we obtain

$$\mathcal{L}(z_k, \lambda^*) = \mathcal{L}(x^*, \lambda^*) + (z_k - x^*)^T \nabla_x \mathcal{L}(x^*, \lambda^*) \tag{12.60}$$
$$+ \tfrac{1}{2}(z_k - x^*)^T \nabla_{xx} \mathcal{L}(x^*, \lambda^*)(z_k - x^*) + o(\|z_k - x^*\|^2).$$

By the complementarity conditions (12.30e) the first term on the right-hand-side of this expression is equal to $f(x^*)$. From (12.30a), the second term is zero. Hence we can rewrite (12.60) as

$$\mathcal{L}(z_k, \lambda^*) = f(x^*) + \tfrac{1}{2}(z_k - x^*)^T \nabla_{xx} \mathcal{L}(x^*, \lambda^*)(z_k - x^*) + o(\|z_k - x^*\|^2). \tag{12.61}$$

By using (12.57) and (12.58), we have for the second-order term and the remainder term that

$$\tfrac{1}{2}(z_k - x^*)^T \nabla_{xx} \mathcal{L}(x^*, \lambda^*)(z_k - x^*) + o(\|z_k - x^*\|^2)$$
$$= \tfrac{1}{2}(t_k/\|w\|)^2 w^T \nabla_{xx} \mathcal{L}(x^*, \lambda^*) w + o(t_k^2).$$

Hence, by substituting this expression together with (12.59) into (12.61), we obtain

$$f(z_k) = f(x^*) + \tfrac{1}{2}(t_k/\|w\|)^2 w^T \nabla_{xx} \mathcal{L}(x^*, \lambda^*) w + o(t_k^2). \tag{12.62}$$

If $w^T \nabla_{xx} \mathcal{L}(x^*, \lambda^*) w < 0$, then (12.62) would imply that $f(z_k) < f(x^*)$ for all k sufficiently large, contradicting the fact that x^* is a local solution. Hence, the condition (12.55) must hold, as claimed. □

Sufficient conditions are conditions on f and c_i, $i \in \mathcal{E} \cup \mathcal{I}$, that ensure that x^* is a local solution of the problem (12.1). (They take the opposite tack to necessary conditions, which assume that x^* is a local solution and deduce properties of f and c_i.) The second-order sufficient condition stated in the next theorem looks very much like the necessary condition just discussed, but it differs in that the constraint qualification is not required, and the inequality in (12.55) is replaced by a strict inequality.

Theorem 12.6 (Second-Order Sufficient Conditions).

Suppose that for some feasible point $x^* \in \mathbb{R}^n$ there is a Lagrange multiplier vector λ^* such that the KKT conditions (12.30) are satisfied. Suppose also that

$$w^T \nabla_{xx} \mathcal{L}(x^*, \lambda^*) w > 0, \quad \text{for all } w \in F_2(\lambda^*), w \neq 0. \tag{12.63}$$

Then x^* is a strict local solution for (12.1).

PROOF. The result is proved if we can show that for any feasible sequence $\{z_k\}$ approaching x^*, we have that $f(z_k) > f(x^*)$ for all k sufficiently large.

Given any feasible sequence, we have from Lemma 12.3(i) and Definition 12.4 that all its limiting directions d satisfy $d \in F_1$. Choose a particular limiting direction d whose associated subsequence $\mathcal{S}_d$ satisfies (12.34). In other words, we have for all $k \in \mathcal{S}_d$ that

$$z_k - x^* = \|z_k - x^*\| d + o(\|z_k - x^*\|). \tag{12.64}$$

From (12.28), we have that

$$\mathcal{L}(z_k, \lambda^*) = f(z_k) - \sum_{i \in \mathcal{A}(x^*)} \lambda_i^* c_i(z_k) \leq f(z_k), \tag{12.65}$$

while the Taylor series approximation (12.61) from the proof of Theorem 12.5 continues to hold.

We know that $d \in F_1$, but suppose first that it is *not* in $F_2(\lambda^*)$. We can then identify some index $j \in \mathcal{A}(x^*) \cap \mathcal{I}$ such that the strict positivity condition

$$\lambda_j^* \nabla c_j(x^*)^T d > 0 \tag{12.66}$$

is satisfied, while for the remaining indices $i \in \mathcal{A}(x^*)$, we have

$$\lambda_i^* \nabla c_i(x^*)^T d \geq 0.$$

From Taylor's theorem and (12.64), we have for all $k \in \mathcal{S}_d$ and for this particular value of j that

$$\begin{aligned} \lambda_j^* c_j(z_k) &= \lambda_j^* c_j(x^*) + \lambda_j^* \nabla c_j(x^*)^T (z_k - x^*) + o(\|z_k - x^*\|) \\ &= \|z_k - x^*\| \lambda_j^* \nabla c_j(x^*)^T d + o(\|z_k - x^*\|). \end{aligned}$$

Hence, from (12.65), we have for $k \in \mathcal{S}_d$ that

$$\begin{aligned} \mathcal{L}(z_k, \lambda^*) &= f(z_k) - \sum_{i \in \mathcal{A}(x^*)} \lambda_i^* c_i(z_k) \\ &\leq f(z_k) - \|z_k - x^*\| \lambda_j^* \nabla c_j(x^*)^T d + o(\|z_k - x^*\|). \end{aligned} \tag{12.67}$$

From the Taylor series estimate (12.61), we have meanwhile that

$$\mathcal{L}(z_k, \lambda^*) = f(x^*) + O(\|z_k - x^*\|^2),$$

and by combining with (12.67), we obtain

$$f(z_k) \geq f(x^*) + \|z_k - x^*\| \lambda_j^* \nabla c_j(x^*)^T d + o(\|z_k - x^*\|).$$

Therefore, because of (12.66), we have $f(z_k) > f(x^*)$ for all $k \in \mathcal{S}_d$ sufficiently large.

For the other case of $d \in F_2(\lambda^*)$, we use (12.64), (12.65), and (12.61) to write

$$\begin{aligned} f(z_k) &\geq f(x^*) + \frac{1}{2}(z_k - x^*)^T \nabla_{xx} \mathcal{L}(x^*, \lambda^*)(z_k - x^*) + o(\|z_k - x^*\|^2) \\ &= \frac{1}{2}\|z_k - x^*\|^2 d^T \nabla_{xx} \mathcal{L}(x^*, \lambda^*) d + o(\|z_k - x^*\|^2). \end{aligned}$$

Because of (12.63), we again have $f(z_k) > f(x^*)$ for all $k \in \mathcal{S}_d$ sufficiently large.

Since this reasoning applies to *all* limiting directions of $\{z_k\}$, and since each element z_k of the sequence can be assigned to one of the subsequences $\mathcal{S}_d$ that converge to one of these limiting directions, we conclude that $f(z_k) > f(x^*)$ for all k sufficiently large. □

❑ **EXAMPLE 12.7** (EXAMPLE 12.2, ONE MORE TIME)

We now return to Example 12.2 to check the second-order conditions for problem (12.17). In this problem we have $f(x) = x_1 + x_2$, $c_1(x) = 2 - x_1^2 - x_2^2$, $\mathcal{E} = \emptyset$, and $\mathcal{I} = \{1\}$. The Lagrangian is

$$\mathcal{L}(x, \lambda) = (x_1 + x_2) - \lambda_1(2 - x_1^2 - x_2^2),$$

and it is easy to show that the KKT conditions (12.30) are satisfied by $x^* = (-1, -1)^T$, with $\lambda_1^* = \frac{1}{2}$. The Lagrangian Hessian at this point is

$$\nabla_{xx} \mathcal{L}(x^*, \lambda^*) = \begin{bmatrix} 2\lambda_1^* & 0 \\ 0 & 2\lambda_1^* \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

This matrix is positive definite, that is, it satisfies $w^T \nabla_{xx} \mathcal{L}(x^*, \lambda^*) w > 0$ for *all* $w \neq 0$, so it certainly satisfies the conditions of Theorem 12.6. We conclude that $x^* = (-1, -1)^T$ is a strict local solution for (12.17). (In fact, it is the global solution of this problem, since, as we note in a later section, this problem is a convex programming problem.)

❑

❑ EXAMPLE 12.8

For an example in which the issues are more complex, consider the problem

$$\min\ -0.1(x_1 - 4)^2 + x_2^2 \qquad \text{s.t.} \qquad x_1^2 + x_2^2 - 1 \geq 0, \tag{12.68}$$

in which we seek to minimize a nonconvex function over the *exterior* of the unit circle. Obviously, the objective function is not bounded below on the feasible region, since we can take the feasible sequence

$$\begin{bmatrix} 10 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 20 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 30 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 40 \\ 0 \end{bmatrix},$$

and note that $f(x)$ approaches $-\infty$ along this sequence. Therefore, no global solution exists, but it may still be possible to identify a strict local solution on the boundary of the constraint. We search for such a solution by using the KKT conditions (12.30) and the second-order conditions of Theorem 12.6.

By defining the Lagrangian for (12.68) in the usual way, it is easy to verify that

$$\nabla_x \mathcal{L}(x, \lambda) = \begin{bmatrix} -0.2(x_1 - 4) - 2\lambda x_1 \\ 2x_2 - 2\lambda x_2 \end{bmatrix}, \tag{12.69a}$$

$$\nabla_{xx} \mathcal{L}(x, \lambda) = \begin{bmatrix} -0.2 - 2\lambda & 0 \\ 0 & 2 - 2\lambda \end{bmatrix}. \tag{12.69b}$$

The point $x^* = (1, 0)^T$ satisfies the KKT conditions with $\lambda_1^* = 0.3$ and the active set $\mathcal{A}(x^*) = \{1\}$. To check that the second-order sufficient conditions are satisfied at this point, we note that

$$\nabla c_1(x^*) = \begin{bmatrix} 2 \\ 0 \end{bmatrix},$$

so that the space F_2 defined in (12.52) is simply

$$F_2(\lambda^*) = \{w \mid w_1 = 0\} = \{(0, w_2)^T \mid w_2 \in \mathbb{R}\}.$$

Now, by substituting x^* and λ^* into (12.69b), we have for any $w \in F_2$ with $w \neq 0$ that

$$w^T \nabla_{xx} \mathcal{L}(x^*, \lambda^*) w = \begin{bmatrix} 0 \\ w_2 \end{bmatrix}^T \begin{bmatrix} -0.4 & 0 \\ 0 & 1.4 \end{bmatrix} \begin{bmatrix} 0 \\ w_2 \end{bmatrix} = 1.4 w_2^2 > 0.$$

Hence, the second-order sufficient conditions are satisfied, and we conclude from Theorem 12.6 that $(1, 0)$ is a strict local solution for (12.68). □

SECOND-ORDER CONDITIONS AND PROJECTED HESSIANS

The second-order conditions are sometimes stated in a form that is weaker but easier to verify than (12.55) and (12.63). This form uses a two-sided projection of the Lagrangian Hessian $\nabla_{xx}\mathcal{L}(x^*, \lambda^*)$ onto subspaces that are related to $F_2(\lambda^*)$.

The simplest case is obtained when the multiplier λ^* that satisfies the KKT conditions (12.30) is unique (as happens, for example, when the LICQ condition holds) and strict complementarity holds. In this case, the definition (12.52) of $F_2(\lambda^*)$ reduces to

$$F_2(\lambda^*) = \text{Null}\left[\nabla c_i(x^*)^T\right]_{i\in\mathcal{A}(x^*)} = \text{Null}\, A,$$

where A is defined as in (12.38). In other words, $F_2(\lambda^*)$ is the null space of the matrix whose rows are the active constraint gradients at x^*. As in (12.40), we can define the matrix Z with full column rank whose columns span the space $F_2(\lambda^*)$. Any vector $w \in F_2(\lambda^*)$ can be written as $w = Zu$ for some vector u, and conversely, we have that $Zu \in F_2(\lambda^*)$ for all u. Hence, the condition (12.55) in Theorem 12.5 can be restated as

$$u^T Z^T \nabla_{xx}\mathcal{L}(x^*, \lambda^*) Zu \geq 0 \quad \text{for all } u,$$

or, more succinctly,

$$Z^T \nabla_{xx}\mathcal{L}(x^*, \lambda^*) Z \quad \text{is positive semidefinite.}$$

Similarly, the condition (12.63) in Theorem 12.6 can be restated as

$$Z^T \nabla_{xx}\mathcal{L}(x^*, \lambda^*) Z \quad \text{is positive definite.}$$

We see at the end of this section that Z can be computed numerically, so that the positive (semi)definiteness conditions can actually be checked by forming these matrices and finding their eigenvalues.

When the optimal multiplier λ^* is unique but the strict complementarity condition is not satisfied, $F_2(\lambda^*)$ is no longer a subspace. Instead, it is an intersection of planes (defined by the first two conditions in (12.52)) and half-spaces (defined by the third condition in (12.52)). We can still, however, define two subspaces $\overline{F}_2$ and $\underline{F}_2$ that "bound" F_2 above and below, in the sense that $\overline{F}_2$ is the smallest-dimensional subspace that *contains* $F_2(\lambda^*)$, while $\underline{F}_2$ is the largest-dimensional subspace *contained in* $F_2(\lambda^*)$. To be precise, we have

$$\underline{F}_2 = \{d \in F_1 \mid \nabla c_i(x^*)^T d = 0, \text{ all } i \in \mathcal{A}(x^*)\},$$

$$\overline{F}_2 = \{d \in F_1 \mid \nabla c_i(x^*)^T d = 0, \text{ all } i \in \mathcal{A}(x^*) \text{ with } i \in \mathcal{E} \text{ or } \lambda_i^* > 0\},$$

so that

$$\underline{F}_2 \subset F_2(\lambda^*) \subset \overline{F}_2. \tag{12.70}$$

As in the previous case, we can construct matrices $\underline{Z}$ and $\overline{Z}$ whose columns span the subspaces $\underline{F}_2$ and $\overline{F}_2$, respectively. If the condition (12.55) of Theorem 12.5 holds, we can be sure that

$$w^T \nabla_{xx} \mathcal{L}(x^*, \lambda^*) w \geq 0, \quad \text{for all } w \in \underline{F}_2,$$

because $\underline{F}_2 \subset F_2(\lambda^*)$. Therefore, an immediate consequence of (12.55) is that the matrix $\underline{Z}^T \nabla_{xx} \mathcal{L}(x^*, \lambda^*) \underline{Z}$ is positive semidefinite.

Analogously, we have from $F_2(\lambda^*) \subset \overline{F}_2$ that condition (12.63) is implied by the condition

$$w^T \nabla_{xx} \mathcal{L}(x^*, \lambda^*) w > 0, \quad \text{for all } w \in \overline{F}_2.$$

Hence, given that the λ^* satisfying the KKT conditions is unique, a sufficient condition for (12.63) is that the matrix $\overline{Z}^T \nabla_{xx} \mathcal{L}(x^*, \lambda^*) \overline{Z}$ be positive definite. Again, this condition provides a practical way to check the second-order sufficient condition.

The matrices $\underline{Z}^T \nabla_{xx} \mathcal{L}(x^*, \lambda^*) \underline{Z}$ and $\overline{Z}^T \nabla_{xx} \mathcal{L}(x^*, \lambda^*) \overline{Z}$ are sometimes called *two-sided projected Hessian* matrices, or simply *projected Hessians* for short.

One way to compute the matrix Z (and its counterparts $\underline{Z}$ and $\overline{Z}$) is to apply a QR factorization to the matrix of active constraint gradients whose null space we seek. In the simplest case above (in which the multiplier λ^* is unique and strictly complementary), we define A as in (12.38) and write the QR factorization of A^T as

$$A^T = Q \begin{bmatrix} R \\ 0 \end{bmatrix} = \begin{bmatrix} Q_1 & Q_2 \end{bmatrix} \begin{bmatrix} R \\ 0 \end{bmatrix} = Q_1 R, \tag{12.71}$$

where R is a square upper triangular matrix, and Q is $n \times n$ orthogonal. If R is nonsingular, we can set $Z = Q_2$. If R is singular (indicating that the active constraint gradients are linearly dependent), a slight enhancement of this procedure that makes use of column pivoting during the QR procedure can be used to identify Z. For more details, see Section 15.2.

CONVEX PROGRAMS

Convex programming problems are constrained optimization problems in which the objective function f is a convex function and the feasible set Ω defined by (12.2) is a convex

set. As for unconstrained optimization (see Chapter 2), the convex case is special—all local solutions are also global solutions, and the set of global minima is itself convex. (The proof of these statements is similar to the proof of Theorem 2.5 and is left as an exercise.) Convex programming problems often can be recognized by the algebraic properties of the constraint functions c_i, as we see in the following theorem.

Theorem 12.7.

A sufficient condition for the feasible region Ω defined by the constraints in (12.1) to be convex is that the functions c_i be linear for $i \in \mathcal{E}$ and that $-c_i$ be convex functions for $i \in I$.

PROOF. Let x_0 and x_1 be two feasible points, that is,

$$\begin{aligned} &c_i(x_0) = c_i(x_1) = 0, \quad i \in \mathcal{E}, \\ &c_i(x_0) \geq 0, \quad c_i(x_1) \geq 0, \quad i \in \mathcal{I}, \end{aligned}$$

and define $x_\tau = (1-\tau)x_0 + \tau x_1$ for any $\tau \in [0, 1]$. To prove the theorem, it is sufficient to show that x_τ is feasible. By linearity of the equality constraints, we have

$$c_i(x_\tau) = (1-\tau)c_i(x_0) + \tau c_i(x_1) = 0, \quad i \in \mathcal{E},$$

so x_τ satisfies the equality constraints. By convexity of $-c_i$ for $i \in I$, we have

$$-c_i(x_\tau) \leq (1-\tau)(-c_i(x_0)) + \tau(-c_i(x_1)) \leq 0, \quad i \in \mathcal{I},$$

so x_τ also satisfies the inequality constraints. Hence $x_\tau \in \Omega$. □

It follows from Theorem 12.7 that linear programming is a special case of convex programming.

Of our examples in this chapter, Examples 12.2, 12.3, and 12.4 were all convex programs, while Example 12.1 was not, because its feasible region was not convex.

12.5 OTHER CONSTRAINT QUALIFICATIONS

We now reconsider the constraint qualification, the "extra" assumption that is needed to make necessary conditions of the KKT conditions of Theorem 12.1. In general, constraint qualifications allow us to show that the set F_1 of Definition 12.4 is identical to the set of multiples of limiting feasible directions. (The LICQ of Definition 12.1 was used in this way in the proof of Lemma 12.3, for instance.) To put it another way, a constraint qualification is a condition that ensures that the linear approximation F_1 to the feasible region at x^* captures the essential geometric features of the true feasible set in some neighborhood of x^*.

This point is perhaps best illustrated by an example in which F_1 contains directions that are obviously infeasible with respect to the actual constraint set. Consider the feasible region defined by

$$x_2 \leq x_1^3, \quad x_2 \geq 0, \tag{12.72}$$

as illustrated in Figure 12.11. Supposing that $x^* = (0, 0)$, so that both constraints are active, we have from Definition 12.4 that

$$F_1 = \{d \mid -d_2 \geq 0, \quad d_2 \geq 0\} = \{d \mid d_2 = 0\}.$$

This set includes the direction $(-1, 0)$, which points along the negative direction of the horizontal axis and is clearly *not* a limiting direction for any feasible sequence. In fact, the only possible limiting direction for this constraint set at $x^* = (0, 0)$ is the vector $(1, 0)$. We conclude that the linear approximation to the feasible set at $(0, 0)$ fails to capture the geometry of the true feasible set, so constraint qualifications do not hold in this case.

One situation in which the linear approximation F_1 is *obviously* an adequate representation of the actual feasible set occurs when all the active constraints are already linear! It is not difficult to prove a version of Lemma 12.3 for this situation.

Lemma 12.8.

Suppose that all active constraints $c_i(\cdot)$, $i \in \mathcal{A}(x^*)$, are linear functions. Then the vector $w \in \mathbb{R}^n$ belongs to F_1 if and only if $w/\|w\|$ is a limiting direction of some feasible sequence.

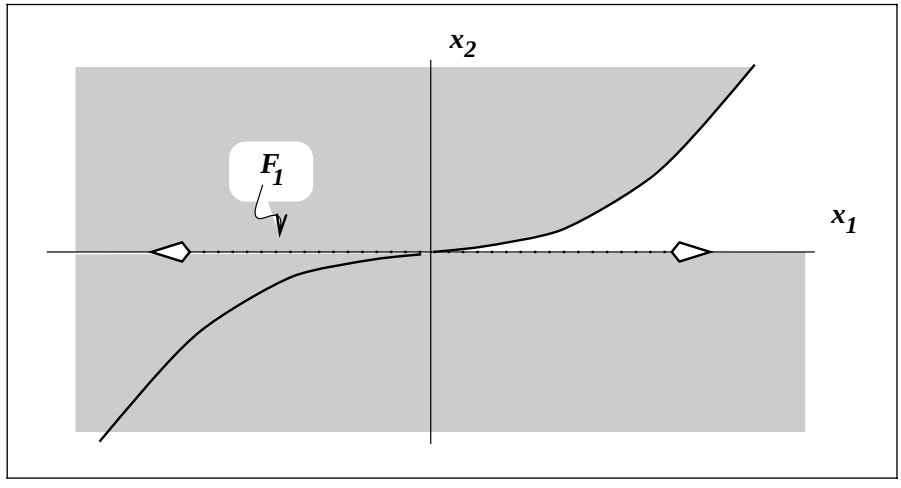


Figure 12.11 Failure of constraint qualifications at $(0, 0)$ for the feasible region defined by (12.72).

PROOF. The reverse implication holds, because Lemma 12.3 shows that all multiples of limiting directions of feasible sequences belong to F_1.

To prove the forward implication, we choose $w \in F_1$ arbitrarily and construct a feasible sequence for which $w/\|w\|$ is a limiting direction. First note that we can choose a positive scalar T such that

$$c_i(x^* + tw) > 0, \quad \text{for all } i \in \mathcal{I}\backslash\mathcal{A}(x^*) \text{ and all } t \in [0, T],$$

that is, the inactive constraints stay inactive at least for short steps along w. Now define the sequence z_k by

$$z_k = x^* + (T/k)w, \quad k = 1, 2, \ldots.$$

Since $a_i^T w \geq 0$ for all $i \in \mathcal{I} \cup \mathcal{A}(x^*)$, we have

$$c_i(z_k) = c_i(z_k) - c_i(x^*) = a_i^T(z_k - x^*) = \frac{T}{k} a_i^T w \geq 0, \quad \text{for all } i \in \mathcal{I} \cap \mathcal{A}(x^*),$$

so that z_k is feasible with respect to the active inequality constraints c_i, $i \in \mathcal{I} \cap \mathcal{A}(x^*)$. By the choice of T, we find that z_k is also feasible with respect to the inactive inequality constraints $i \in \mathcal{I}\backslash\mathcal{A}(x^*)$, and it is easy to show that $c_i(z_k) = 0$ for the equality constraints $i \in \mathcal{E}$. Hence, z_k is feasible for each $k = 1, 2, \ldots$. In addition, we have that

$$\frac{z_k - x^*}{\|z_k - x^*\|} = \frac{(T/k)w}{(T/k)\|w\|} = \frac{w}{\|w\|},$$

so that indeed the direction $w/\|w\|$ is the limiting feasible direction of the sequence $\{z_k\}$. □

We conclude from this result that the condition that all active constraints be linear is another possible constraint qualification. It is neither weaker nor stronger than the LICQ condition, that is, there are situations in which one condition is satisfied but not the other (see the exercises).

Another useful generalization of the LICQ is the Mangasarian–Fromovitz constraint qualification (MFCQ), which we define formally as follows.

Definition 12.5 (MFCQ).

Given the point x^* and the active set $\mathcal{A}(x^*)$ defined by (12.29), we say that the Mangasarian–Fromovitz constraint qualification (MFCQ) *holds if there exists a vector $w \in \mathbb{R}^n$ such that*

$$\begin{aligned} \nabla c_i(x^*)^T w &> 0, \quad \text{for all } i \in \mathcal{A}(x^*) \cap \mathcal{I}, \\ \nabla c_i(x^*)^T w &= 0, \quad \text{for all } i \in \mathcal{E}, \end{aligned}$$

and the set of equality constraint gradients $\{\nabla c_i(x^*), i \in \mathcal{E}\}$ *is linearly independent.*

Note the *strict* inequality involving the active inequality constraints.

The MFCQ is a weaker condition than LICQ. If LICQ is satisfied, then the system of equalities defined by

$$\begin{aligned} \nabla c_i(x^*)^T w &= 1, \quad \text{for all } i \in \mathcal{A}(x^*) \cap \mathcal{I}, \\ \nabla c_i(x^*)^T w &= 0, \quad \text{for all } i \in \mathcal{E}, \end{aligned}$$

has a solution w, by full rank of the active constraint gradients. Hence, we can choose the w of Definition 12.5 to be precisely this vector. On the other hand, it is easy to construct examples in which the MFCQ is satisfied but the LICQ is not; see the exercises.

It is possible to prove a version of the first-order necessary condition result, Theorem 12.1, in which MFCQ replaces LICQ in the assumptions. MFCQ has the particularly nice property that it is equivalent to boundedness of the set of Lagrange multiplier vectors λ^* for which the KKT conditions (12.30) are satisfied. (In the case of LICQ, this set consists of a unique vector λ^*, and so is trivially bounded.)

Note that constraint qualifications are *sufficient conditions* for the linear approximation to be adequate, not necessary conditions. For instance, consider the set defined by $x_2 \geq -x_1^2$ and $x_2 \leq x_1^2$ and the feasible point $x^* = (0, 0)$. None of the constraint qualifications we have discussed are satisfied, but the linear approximation $F_1 = \{w \mid w_2 = 0\}$ accurately reflects the geometry of the feasible set near x^*.

12.6 A GEOMETRIC VIEWPOINT

Finally, we mention an alternative first-order optimality condition that depends only on the geometry of the feasible set Ω and not on its particular algebraic description in terms of the functions c_i, $i \in \mathcal{E} \cup \mathcal{I}$. In geometric terms, our problem (12.1) can be stated as

$$\min \ f(x) \quad \text{subject to } x \in \Omega, \tag{12.73}$$

where Ω is the feasible set defined by the conditions (12.2).

The tools we need to state this condition are discussed in the Appendix; see in particular the Definition A.1 for the *tangent cone* $T_\Omega(x^*)$ and formula (A.22), which defines the *normal cone* $N_\Omega(x^*)$.

The first-order necessary condition for (12.73) is delightfully simple.

Theorem 12.9.

Suppose that x^* *is a local minimizer of* f *in* Ω. *Then*

$$-\nabla f(x^*) \in N_\Omega(x^*). \tag{12.74}$$

PROOF. Given any $d \in T_\Omega(x^*)$, we have from Definition A.1 that for any decreasing positive sequence $t_j \downarrow 0$, there is a sequence of vectors d_j such that

$$x^* + t_j d_j \in \Omega, \quad d_j \to d. \tag{12.75}$$

(We simply take $x_j \equiv x^*$ in Definition A.1.) Since x^* is a local solution, we must have

$$f(x^* + t_j d_j) \geq f(x^*)$$

for all j sufficiently large. Hence, since f is continuously differentiable, we have from Taylor's theorem (2.4) that

$$t_j \nabla f(x^*)^T d_j + o(t_j) \geq 0.$$

By taking the limits of this inequality as $j \to \infty$, we have

$$\nabla f(x^*)^T d \geq 0.$$

Recall that d was an arbitrary member of $T_\Omega(x^*)$, so we have $-\nabla f(x^*)^T d \leq 0$ for all $d \in T_\Omega(x^*)$. We conclude from (A.22) that $-\nabla f(x^*) \in N_\Omega(x^*)$. □

This result, together with some observations we made earlier in this chapter, suggests that there is a close relationship between $N_\Omega(x^*)$ and the convex combination of active constraint gradients given by (12.47). When the linear independence constraint qualification holds, we can prove this result formally by using an argument based on the implicit function theorem.

Lemma 12.10.

Suppose that the LICQ assumption (Definition 12.1) holds at x^*. Then the tangent cone $T_\Omega(x^*)$ is identical to F_1 from Definition 12.4, while the normal cone $N_\Omega(x^*)$ is simply $-N$, where N is the set defined in (12.47).

PROOF. Suppose that $d \in T_\Omega(x^*)$. Then by Definition A.1, we have for any positive decreasing sequence $t_j \downarrow 0$ that there is a sequence $\{d_j\}$ with the property (12.75). In particular,

$$c_i(x^* + t_j d_j) = 0, \quad i \in \mathcal{E}, \qquad c_i(x^* + t_j d_j) \geq 0, \quad i \in \mathcal{A}(x^*) \cap \mathcal{I}.$$

For $i \in \mathcal{E}$, we have by Taylor's theorem that

$$0 = c_i(x^*) + t_j a_i^T d_j + o(t_j) = t_j a_i^T d_j + o(t_j).$$

By taking the limit as $j \to \infty$, we obtain $a_i^T d = 0$. Similarly, we obtain $a_i^T d \geq 0$ for $i \in \mathcal{A}(x^*) \cap \mathcal{I}$. Hence $d \in F_1$.

For the converse, choose $d \in F_1$ and let x_j and t_j be given sequences as in Definition A.1. For simplicity, we assume as in Lemma 12.3 that all constraints c_i, $i = 1, 2, \ldots, m$, are active at the solution. Similarly to (12.41), and using the notation

$$A(x)^T = [\nabla c_i(x)]_{i=1,2,\ldots,m}, \quad A^* = A(x^*), \quad A_j = A(x_j),$$

we construct the following parametrized system of equations:

$$R(z, t; x_j) = \begin{bmatrix} c(z) - c(x_j) - tA_j d \\ Z^T(z - x_j - td) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \tag{12.76}$$

Clearly, the solution for $t = 0$ is $z(0) = x_j$. Further, we have

$$\nabla_z R(z, t; x_j)\big|_{t=0} = \begin{bmatrix} A_j \\ Z \end{bmatrix},$$

which is nonsingular for j sufficiently large, since $A(x_j) \to A(x^*)$, so the solution $z(t)$ of (12.76) is well-defined for all values of t sufficiently small. Moreover, from Definition 12.4, we have for such solutions z that

$$\begin{aligned} c_i(z(t)) &= c(x_j) + tA_j d = 0 \quad \text{for } i \in \mathcal{E}; \\ c_i(z(t)) &= c(x_j) + tA_j d \geq 0 \quad \text{for } i \in \mathcal{A} \cap \mathcal{I}, \end{aligned}$$

so that $z(t)$ is feasible with respect to the constraints in (12.2).

We now define

$$d_j = \frac{z(t_j) - x_j}{t_j}.$$

Since $x_j + t_j d_j = z(t_j) \in \Omega$, the sequence $\{d_j\}$ satisfies one of the properties of Definition A.1. To verify the other property, we rearrange (12.76) to obtain

$$\begin{aligned} \begin{bmatrix} A_j \\ Z^T \end{bmatrix} d &= \frac{1}{t_j} \begin{bmatrix} c(z(t_j)) - c(x_j) \\ z(t_j) - x_j \end{bmatrix} \\ &= \begin{bmatrix} A_j \\ Z^T \end{bmatrix} d_j + \frac{1}{t_j} O(\|c(z(t_j)) - c(x_j) - t_j A_j d_j\|). \end{aligned} \tag{12.77}$$

By using Taylor's theorem, we have that

$$c(z(t_j)) - c(x_j) = A_j(z(t_j) - x_j) + o(\|z(t_j) - x_j\|)$$

$$= A^*(z(t_j) - x_j) + (A_j - A^*)(z(t_j) - x_j) + o(\|z(t_j) - x_j\|)$$
$$= A^*(z(t_j) - x_j) + o(\|z(t_j) - x_j\|),$$

and therefore from the definition of d_j, we have

$$\frac{1}{t_j}[c(z(t_j)) - c(x_j) - t_j A^* d_j] = o(\|z(t_j) - x_j\|/t_j) = o(\|d_j\|).$$

By substituting into (12.77), we obtain

$$\begin{bmatrix} A^* \\ Z^T \end{bmatrix} d = \begin{bmatrix} A^* \\ Z^T \end{bmatrix} d_j + o(\|d_j\|),$$

which implies, by nonsingularity of the matrix $[(A^*)^T \mid Z]$, that

$$d = d_j + o(\|d_j\|).$$

Hence, we have $d_j \to d$, and so $d \in T_\Omega(x^*)$, completing the proof of equivalence between F_1 and $T_\Omega(x^*)$.

The proof of equivalence between $-N$ and $N_\Omega(x^*)$ follows from Lemma 12.4 and the definition (A.22) of $N_\Omega(x^*)$. From Lemma 12.4, we have that

$$g \in N \quad \Rightarrow \quad g^T d \geq 0 \ \text{ for all } d \in F_1.$$

Since we have $F_1 = T_\Omega(x^*)$, it follows by switching the sign of this expression that

$$g \in -N \quad \Rightarrow \quad g^T d \leq 0 \ \text{ for all } d \in T_\Omega(x^*).$$

By comparing with (A.22), we see that this is precisely the definition of $N_\Omega(x^*)$, so we conclude that $N_\Omega(x^*) = -N$, as claimed. □

NOTES AND REFERENCES

We return to our claim that the set N defined by (12.47) is a closed set. This fact is needed in the proof of Lemma 12.4 to ensure that the solution of the projection subproblem (12.48) is well-defined. Note that N has the following form:

$$N = \{s \mid s = A\lambda, \ C\lambda \geq 0\}, \tag{12.78}$$

for appropriately defined matrices A and C.

We outline a proof of closedness of N for the case in which LICQ holds (Definition 12.1), that is, the matrix A has full column rank. It suffices to show that whenever $\{s_k\}$ is

a sequence of vectors satisfying $s_k \to s^*$, then $s^* \in N$. Because A has full column rank, for each s_k there is a unique vector λ_k such that $s_k = A\lambda_k$; in fact, we have $\lambda_k = (A^T A)^{-1} A^T s_k$. Because $s_k \in N$, we must have $C\lambda_k \geq 0$, and since $s_k \to s^*$, we also have that

$$\lim_k \lambda_k = \lim_k (A^T A)^{-1} A^T s_k = (A^T A)^{-1} A^T s^* \stackrel{\text{def}}{=} \lambda^*.$$

The inequality $C\lambda^* \geq 0$ is a consequence of $C\lambda_k \geq 0$ for all k, whereas $s^* = A\lambda^*$ follows from the facts that $s_k - A\lambda_k = 0$ for all k, and $s_k - A\lambda_k \to s^* - A\lambda^*$. Therefore we have identified a vector λ^* such that $s^* = A\lambda^*$ and $C\lambda^* \geq 0$, so $s^* \in N$, as claimed.

There is a more general proof, using an argument due to Mangasarian and Schumaker [155] and appealing to Theorem 13.2 (iii), that does not require LICQ. We omit the details here.

The theory of constrained optimization is discussed in many books on numerical optimization. The discussion in Fletcher [83, Chapter 9] is similar to ours, though a little terser, and includes additional material on duality. Bertsekas [9, Chapter 3] emphasizes the role of duality and discusses sensitivity of the solution with respect to the active constraints in some detail. The classic treatment of Mangasarian [154] is particularly notable for its thorough description of constraint qualifications.

The KKT conditions were described in a 1951 paper of Kuhn and Tucker [145], though they were derived earlier (and independently) in an unpublished 1939 master's thesis of W. Karush. Lagrange multipliers and optimality conditions for general problems (including nonsmooth problems) are described in the article of Rockafellar [217], which is both wide-ranging and deep.

EXERCISES

12.1 Does problem (12.4) have a finite or infinite number of local solutions? Use the first-order optimality conditions (12.30) to justify your answer.

12.2 Show by drawing a picture that the sufficient conditions for Ω to be convex in Theorem 12.7 are not necessary. That is, it is possible to have a convex feasible region Ω even if the constraint functions do not satisfy the conditions of the theorem.

12.3 If f is convex and the feasible region Ω is convex, show that local solutions of the problem (12.3) are also global solutions. Show that the set of global solutions is convex.

12.4 Let $v : \mathbb{R}^n \to \mathbb{R}^m$ be a smooth vector function and consider the unconstrained optimization problems of minimizing $f(x)$ where

$$f(x) = \|v(x)\|_\infty, \qquad f(x) = \max_{i=1,2,\ldots,m} v_i(x).$$

Reformulate these (generally nonsmooth) problems as smooth constrained optimization problems.

✎ * **12.5** Can you perform a smooth reformulation as in the previous question when f is defined by

$$f(x) = \min_{i=1,2,\ldots,m} f_i(x)?$$

(N.B. "min" not "max.") Why or why not?

✎ **12.6** We demonstrated in this chapter how a feasible region with nonsmooth boundary could sometimes be described by a set of smooth constraints. Conversely, it is possible that a single smooth constraint describes a feasible set with nonsmooth boundary. As an example, consider the cone defined by

$$\left\{(x_1, x_2, x_3) \,|\, x_3^2 \geq x_1^2 + x_2^2\right\}$$

(draw this set). Can you think of other examples?

✎ **12.7** Show that the vector defined by (12.14) satisfies both (12.12) and (12.13) when the first-order optimality condition (12.10) is not satisfied.(Hint: Use the Hölder inequality (A.37).)

✎ **12.8** Verify that for the sequence $\{z_k\}$ defined by (12.35), the function $f(x) = x_1 + x_2$ satisfies $f(z_{k+1}) > f(z_k)$ for $k = 2, 3, \ldots$. (Hint: Consider the trajectory $z(s) = (-\sqrt{2 - 1/s^2}, -1/s)^T$ and show that the function $h(s) \stackrel{\text{def}}{=} f(z(s))$ has $h'(s) > 0$ for all $s \geq 2$.)

✎ **12.9** Consider the problem (12.9). Specify *two* feasible sequences that approach the maximizing point (1, 1), and show that neither sequence is a decreasing sequence for f.

✎ **12.10** Verify that neither the LICQ nor the MFCQ holds for the constraint set defined by (12.72) at $x^* = (0, 0)$.

✎ **12.11** Consider the feasible set in $\mathbb{R}^2$ defined by $x_2 \geq 0$, $x_2 \leq x_1^2$. Does the linear approximation F_1 of Definition 12.4 adequately capture the geometry of the feasible set near $x^* = (0, 0)$? Is the LICQ or MFCQ satisfied?

✎ **12.12** It is trivial to construct an example of a feasible set and a feasible point x^* at which the LICQ is satisfied but the constraints are nonlinear. Give an example of the reverse situation, that is, where the active constraints are linear but the LICQ is not satisfied.

✎ **12.13** Show that for the feasible region defined by

$$(x_1 - 1)^2 + (x_2 - 1)^2 \leq 2,$$
$$(x_1 - 1)^2 + (x_2 + 1)^2 \leq 2,$$

$$x_1 \geq 0,$$

the MFCQ is satisfied at $x^* = (0, 0)$ but the LICQ is not satisfied.

✎ **12.14** Verify (12.70).

✎ **12.15** Consider the half space defined by $H = \{x \in \mathbb{R}^n \mid a^T x + \alpha \geq 0\}$ where $a \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$ are given. Formulate and solve the optimization problem for finding the point x in H that has the smallest Euclidean norm.

✎ **12.16** (Fletcher [83]) Solve the problem

$$\min_x \; x_1 + x_2 \;\; \text{subject to } x_1^2 + x_2^2 = 1$$

by eliminating the variable x_2. Show that the choice of sign for a square root operation during the elimination process is critical; the "wrong" choice leads to an incorrect answer.

✎ **12.17** Prove that when the KKT conditions (12.30) and the LICQ are satisfied at a point x^*, the Lagrange multiplier λ^* in (12.30) is unique.

✎ **12.18** Consider the problem of finding the point on the parabola $y = \frac{1}{5}(x-1)^2$ that is closest to $(x, y) = (1, 2)$, in the Euclidean norm sense. We can formulate this problem as

$$\min \; f(x, y) = (x-1)^2 + (y-2)^2 \quad \text{subject to } (x-1)^2 = 5y.$$

(a) Find all the KKT points for this problem. Is the LICQ satisfied?

(b) Which of these points are solutions?

(c) By directly substituting the constraint into the objective function and eliminating the variable x, we obtain an unconstrained optimization problem. Show that the solutions of this problem cannot be solutions of the original problem.

✎ **12.19** Consider the problem

$$\min_{x \in \mathbb{R}^2} f(x) = -2x_1 + x_2 \quad \text{subject to } \begin{cases} (1-x_1)^3 - x_2 & \geq & 0 \\ x_2 + 0.25x_1^2 - 1 & \geq & 0. \end{cases}$$

The optimal solution is $x^* = (0, 1)^T$, where both constraints are active. Does the LICQ hold at this point? Are the KKT conditions satisfied?

✎ **12.20** Find the minima of the function $f(x) = x_1 x_2$ on the unit circle $x_1^2 + x_2^2 = 1$. Illustrate this problem geometrically.

✎ **12.21** Find the *maxima* of $f(x) = x_1 x_2$ over the unit disk defined by the inequality constraint $1 - x_1^2 - x_2^2 \geq 0$.